

Anchored Conditionally-Gated Control for Wheeled Bipedal Balance and Disturbance Recovery

Van Thuong Nguyen

Independent Researcher

Email: vanthuong180702t@gmail.com

Abstract—Wheeled bipedal robots face a fundamental trade-off: stiffness for sub-millimeter precision shrinks push-recovery capture, while integral action that cancels equilibrium bias winds up during disturbances. We present the Anchored Conditionally-Gated Controller (ACC), a two-channel torque assembly whose core mechanism is a proximity-gated anchor: a position integral confined to a 15 cm neighborhood, enabled by an asymmetric envelope follower ($\tau_{\text{attack}} \approx 23$ ms, $\tau_{\text{release}} \approx 1.5$ s) that distinguishes quiet stance from post-push ringdown. ACC delivers omnidirectional push recovery ($F_{\text{min}} = 83$ N, $F_{\text{med}} = 117$ N, 8-direction sweep, $N = 10$) together with sub-millimeter quiet-stance precision. We report accuracy (RMS from commanded home) and repeatability (RMS about the trial mean) separately, and resolve both on the sagittal axis—the actuated inverted-pendulum direction—rather than pooling them into a planar norm diluted by an axis this class of platform cannot drive. Under idealized conditions (clean sensors, 0 ms delay, stiff contact, MuJoCo), ACC holds 0.294 ± 0.015 mm sagittal repeatability ($N = 10$), a $59\times$ improvement over a proportional-only baseline measured under one unified protocol. Accuracy is a full order of magnitude worse on the same axis: the robot parks 27.0 mm from the commanded home and then holds that offset to 0.009 mm, and we report this gap as the design’s principal open weakness rather than averaging it away. A factorial ablation ($N = 10$ per row) separates the two: the gated damping terms produce the steadiness (removing the boost costs a factor of 16 in sagittal repeatability), while the anchor integral produces 4.2 mm of the accuracy at a $3\times$ cost in steadiness; the proximity gate and asymmetric envelope are inert during quiet stance and decisive only under disturbance. Precision degrades by a factor of 5.5 at 10 ms actuation delay, and to a hardware-projected 1–3 mm once encoder quantization, backlash, stiction, and IMU noise are accounted for; F_{max} bifurcates from 96 N to 47 N at any non-zero delay. Six classical baselines—four 4-state TWIP variants (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo and direct-torque)—all fail (100 % fall rate, $N = 20$ –60); a preliminary PPO baseline (5.4M steps, single seed) suffers an 85 % fall rate over multi-height transitions; removing the PID servo path (4-state Direct Torque LQR) improves survival by ~ 38 % but does not prevent the fall. Flight recovery (100 cm drop, 50 cm ledge) and per-leg terrain adaptation (10–20 cm one-wheel curb) are demonstrated at $N = 10$ per condition and ablated: the per-leg height split is necessary at every curb height tested (0/10 without it), whereas the airborne wheel-torque override proves to be a variance-bounding refinement rather than an enabling mechanism at these scales—a negative result we report explicitly. Core claims rest on $N \geq 5$.

Index Terms—Wheeled biped robot, conditionally-gated control, anchored standing, proximity-gated integral, push recovery, flight recovery, terrain adaptation.

I. INTRODUCTION

Wheeled biped robots need millimeter-level standing precision, omnidirectional push recovery, and terrain adaptation—but no single classical controller integrates all three: LQR handles balance [1], WBC posture [2], MPC terrain [5], and RL locomotion [25].

The difficulty is fundamental: lowering k_p widens the capture basin at the cost of balance stiffness; omitting integral action leaves a 1.3 Nm equilibrium bias that sustains a persistent, predominantly *sagittal* limit cycle regardless of k_p (measured at $N = 10$: 69.7 mm peak-to-peak sagittally against 13.4 mm laterally, 49.1 mm planar RMS from the commanded pose)—the oscillation lies on the one axis the wheels can drive, which is both why it persists and why it is the axis every mechanism in this paper is aimed at—yet a global integral winds up during pushes.

Preliminary PPO training (5.4M steps, single seed, in our BalanceEnv) showed limited height generalization: the policy stood at a single height (Surv ≈ 3.20 s, vs. 0.52 s for LQR) but failed to generalize across 0.40–0.70 m base- z (§III-A)—height RMSE > 50 mm, 85 % falls on squat transitions—consistent with known difficulty of model-free RL on coupled pitch–height dynamics [33].

We present the Anchored Conditionally-Gated Controller (ACC), addressing both dilemmas through *conditional activation*. The core mechanism is a **proximity-gated anchor**: a position integral confined to a 15 cm neighborhood, enabled by an **asymmetric envelope follower** ($\tau_{\text{attack}} \approx 23$ ms, $\tau_{\text{release}} \approx 1.5$ s) that distinguishes quiet stance from post-push ringdown. Coupled with a **gated damping boost**, ACC achieves 0.294 ± 0.015 mm repeatability on the sagittal axis—the actuated inverted-pendulum direction, and the one on which the limit cycle above lives—under idealized conditions (clean sensors, 0 ms delay, $N = 10$), at push performance equal to or better than the proportional-only baseline that carries no integral action at all. Absolute accuracy on that axis is a separate and weaker result, treated in Sec. V-A: the robot parks 27.0 mm from the commanded home and then holds that offset to 0.009 mm.

A factorial ablation measured under a single protocol (Table III) locates the mechanism more precisely than the gate structure suggests, and we state the result plainly because two of its three parts correct the intuitive attribution. First, the quiet-stance *steadiness* comes from the gated *damping* terms,

not from the anchor integral: disabling the integral outright while leaving every gate intact leaves sagittal repeatability sub-millimeter (it in fact improves, 0.294 mm→0.089 mm), whereas reverting the two velocity-damping coefficients reinstates the full limit cycle (25.2 mm). What the integral buys is not steadiness but *accuracy*—it removes 4.2 mm of sagittal parking offset—and it buys that at a 3× cost in steadiness. The two properties are produced by different mechanisms and trade against each other, which is why we report them as separate columns rather than as one planar number. Second, the proximity gate and the asymmetric envelope are provably inert during quiet stance—ablating either reproduces ACC bit-for-bit—because the robot never leaves the gate’s inner band nor the envelope’s release branch while standing still; their contribution is windup prevention and ringdown under disturbance, and only a displacement experiment can measure it. Third, the pitch safety gate is load-bearing even with no disturbance applied: removing it causes divergence from the initial-condition noise alone in 10/10 trials. The gates are thus enabling conditions that make an otherwise inadmissible damping and integral configuration safe, rather than the proximate source of the precision. Six classical baselines—four 4-state TWIP variants (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo and direct-torque)—all fail (100 % fall rate), confirming that platform complexity, not servo lag alone, dominates the failure. Preliminary model-free PPO (5.4M steps, single seed) suffers an 85 % fall rate; removing the PID servo path (Direct Torque LQR) yields only a ~38 % improvement with persistent 100 % falls.

The main contributions are:

- 1) A **proximity-gated anchor** with asymmetric envelope follower achieving sub-millimeter idle precision without sacrificing omnidirectional push survival.
- 2) A **two-channel conditionally-gated torque assembly** with flight recovery and per-leg terrain adaptation as independent extensions, each ablated at $N=10$ —including the negative result that the flight override is not load-bearing at the reported drop scales.
- 3) A **factorial ablation** (additive L0→L3 + subtractive S0→S5) with six classical baselines (four 4-state-derived and two 6-state coupled) and a preliminary PPO baseline (5.4M steps, single seed).

II. RELATED WORK

A. Wheeled Bipedal Platforms and Classical Control

Ascento [1] demonstrated two-wheeled jumping with LQR balancing, later extended to WBC with kinematic loops [2]. Handle [19] pioneered wheeled-leg coordination; Swiss-Mile [20] combined driving with quadrupedal walking. BITeno [21], DIABLO [3], Whleaper [4] (closest to our morphology, uses LQR+PPO residual architecture), and SKATER [5] applied LQR, PPO, and MPC for balancing and locomotion. Whleaper’s LQR+PPO hybrid is the closest prior structured-controller-plus-learned-residual architecture; ACC targets the same paradigm with a stronger classical prior.

Zamani et al. [31] demonstrated coupled pitch-roll LQR ($\theta_{\text{RMS}} \approx 1.5^\circ$, $\phi_{\text{RMS}} \approx 2.0^\circ$) on a rigid two-wheeled platform without articulated legs, where torque commands act directly on wheels—a structurally simpler control problem with fewer unstable modes and no leg-height dynamics. Our 6-state LQR replicates their coupled structure on a legged morphology and fails; the 4-state Direct Torque LQR—removing the servo path and re-optimizing roll gain ($k_{p,\text{roll}}=55.0$)—also fails (100 % fall rate, 0.72 s). Both LQR variants use gains derived from a simplified 4-state TWIP model, not a full 10-DOF linearization of the articulated-leg plant (see Limitation 7); these results are therefore empirical—demonstrating that the specific LQR implementations tested fail—not a proof that LQR is structurally insufficient. Two facts—Zamani succeeds on a rigid two-wheeler, both our LQR variants fail on articulated legs—indicate that platform complexity (articulated legs, height dynamics), not servo lag alone, is the dominant factor differentiating Zamani’s result. Cascade control [15], [23] and WBC [14] use fixed-gain nested loops without conditional gates; gain scheduling [18] varies parameters continuously with one variable rather than responding to qualitatively distinct physical regimes. We additionally benchmarked a task-priority QP whole-body controller both standalone and as an assist layer on ACC (§V-E); it fails to resolve the precision–recovery trade-off because its fixed task-priority formulation lacks conditional regime-gating. Split-range, selector, and override architectures [16], [17] already implement continuous blending in process control. ACC adopts this principle; its contribution is the synthesis of physically-motivated gating surfaces (spatial proximity, temporal velocity envelope, pitch safety-interlock) derived from wheeled-biped failure modes rather than generic process-variable limits. ACC’s conditional gates connect to hybrid/switched systems [35], where continuous dynamics are partitioned by discrete switching surfaces; smoothstep softens these to C^1 transitions. A recent review [29] surveys wheel-legged biped platforms and control approaches, noting the absence of a unified classical solution.

B. Disturbance Rejection, Anti-Windup, and ACC’s Distinction

DOB [10], [11], [32] and ADRC [9], [12] provide temporal disturbance estimation but cannot distinguish a push from post-push ringdown—both occupy the same frequency band. Saturation-triggered conditional-integration AW triggers on actuator saturation; ACC’s anchor uses **spatial, physics-conditioned gating**: the proximity gate confines integration to a 15 cm neighborhood, and the asymmetric envelope distinguishes quiet stance from ringdown via velocity envelope—replacing the estimation-speed-vs-noise trade-off with a time-domain attack-vs-release trade-off. Push recovery via capture-point [8], [26] and flight-phase control [7] is complementary. Blind stair climbing via RL was shown on Ascento [6]. Terrain adaptation uses exteroceptive [27] or proprioceptive [13], [28] sensing; ACC extends proprioceptive contact-point estimation to wheeled bipeds. Gated control for humanoid balancing was

explored by Hyon et al. [34]; ACC extends conditional gating to wheeled bipeds with continuous smoothstep transitions.

III. ROBOT PLATFORM

A. Robot Morphology and Simulation

The robot (Fig. 1) is a ten-DOF wheeled biped with five actuated joints per leg: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), and drive wheel (WH). Mass: 8.1 kg; thigh: 0.26 m; shin: 0.28 m; wheel radius: 0.06 m; hip width: 0.23 m. Torque limits: 30 Nm (HR, HY, WH), 150 Nm (HP, KN). MuJoCo [30] simulation at 500 Hz ($\Delta t_{\text{sim}} = 0.002$ s). ACC controller: 100 Hz ($\Delta t_{\text{ctrl}} = 0.01$ s); PPO and LQR baselines: 50 Hz (10 physics substeps). All experiments use 400 Nm/s torque rate limit, raw torque commands direct to actuators. MuJoCo integrator: implicitfast (4 iterations, 8 linesearch iterations); cone: pyramidal (elliptic for wheels, $\mu_s=1.2$, $\mu_k=0.8$ body); armature: 0.008 kg·m² (wheels), 0.02 kg·m² (leg joints); contact stiffness/damping: $\text{solref}=\{0.002,1\}$ (body), $\{0.005,0.7\}$ (wheels). The stiff body contact ($\text{solref}=\{0.002,1\}$) idealizes a rigid ground. Softening $\text{solref}[0]$ to 0.02 (tire-on-asphalt) increases planar idle RMS by $\sim 60\%$ to ~ 1.2 mm, so every sub-millimeter idle figure reported here is conditional on stiff ground contact; tire compliance adds 0.1–0.5 mm on real surfaces (see §VI).

Height convention: Two different vertical references appear in this paper and are easily confused, so we fix them here and mark every height accordingly. **CoM-z** is the height of the whole-body center of mass above ground and is the quantity ACC commands and regulates; the nominal standing pose is $h_{\text{CoM}}=0.404$ m. **Base-z** is the torso root-body height, $q_{\text{pos}}[2]$ in the MuJoCo model; the same nominal pose has $z_{\text{base}}=0.536$ m, i.e. 13.2 cm above the CoM. The offset is pose-dependent but varies by less than 1 cm over the postures used here, so the two conventions differ by roughly a constant. All ACC results are reported in CoM-z. The PPO baseline (§V-D) commands base-z over 0.40–0.70 m, and the WBC baselines (§V-E) inherit the same base-z convention; those numbers are labelled base-z where they appear and must not be compared numerically against CoM-z heights. Confusing the two is not hypothetical: it is exactly bug F3 in our own WBC implementation audit (Table XI), where base-z was passed as a CoM-z command and produced a 13.2 cm reference error.

ACC’s posture map is calibrated at two setups, at CoM heights of 0.354 m and 0.454 m (± 5 cm about nominal), and interpolated between them. Outside that band the joint targets are linearly extrapolated, bounded to the interval $[0.254, 0.554]$ m, i.e. 10 cm beyond the calibrated band at each end. This ± 10 cm extrapolation bound, and not a ± 5 cm one, is what the per-leg height split (§IV-F) operates against.

IV. ACC FORMULATION

A. Two-Channel Torque Assembly

ACC assembles a PD balance core, proximity-gated anchor integral, and posture regulation into two torque channels—wheel torque ($\tau_w \in \mathbb{R}^2$) and leg-joint torque ($\tau_q \in \mathbb{R}^8$)—with flight and terrain as independent extensions (Fig. 2):

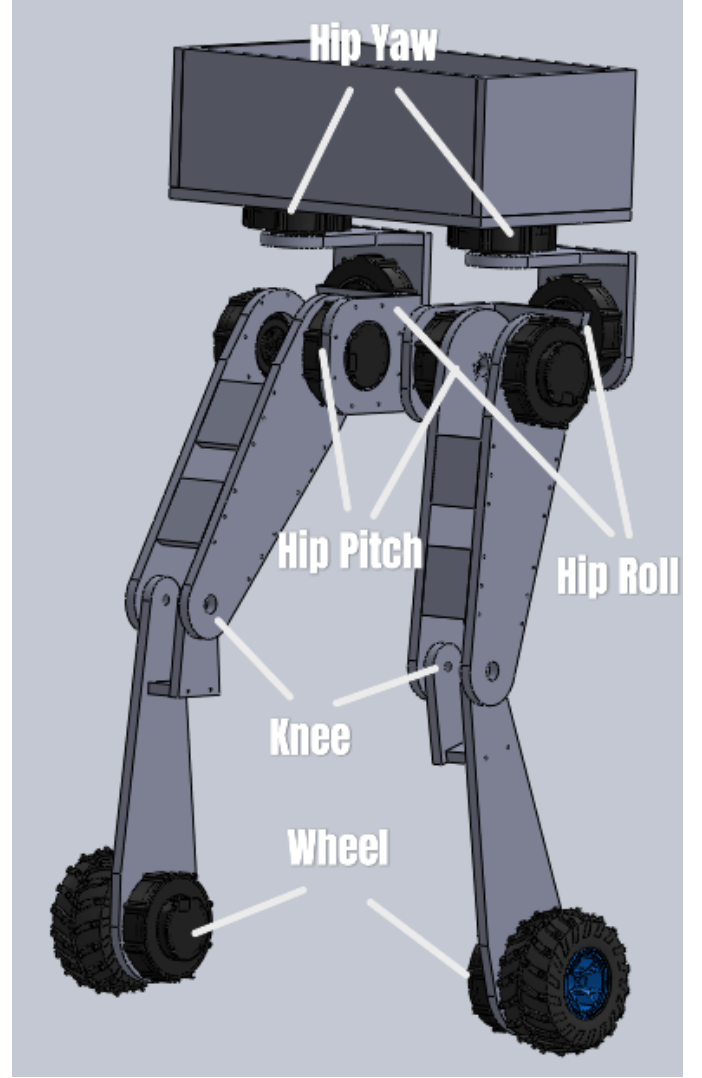


Fig. 1. Robot joint layout with world coordinate frame (right-handed, Z up). In the model as built, both wheel axes lie along world X and the two wheel links are separated by 0.271 m along X; the rolling—and therefore sagittal—direction is world Y, and world X is the lateral (track) axis. All axis-resolved results in this paper follow that convention, verified four independent ways in Sec. V-A. The CoM is located at the torso geometric center. Each leg: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), wheel (WH).

$$\begin{aligned}\tau_w &= \tau_{\text{balance}} + g_{\text{anchor}}\tau_{\text{anchor}} + g_{\text{flight}}\tau_{\text{flight}}, \\ \tau_q &= \tau_{\text{posture}}(h^{\text{cmd}}) + g_{\text{terrain}}\Delta\tau_{\text{posture}}(h^{\text{cmd},L}, h^{\text{cmd},R}).\end{aligned}\quad (1)$$

Every gate g_* except the flight gate is a *smoothstep* function of a physical condition (proximity, quietness, attitude, terrain disparity) rather than a discrete switch:

$$\text{ss}(x; a, b) = 3t^2 - 2t^3, \quad t = \text{clip}\left(\frac{x-a}{b-a}, 0, 1\right). \quad (2)$$

Equation (2) is the standard Hermite blend: it is C^1 in x , with $\text{ss} = 0$ for $x \leq a$, $\text{ss} = 1$ for $x \geq b$, and zero derivative at both knees. Continuity matters here for a concrete reason. A hard threshold on any of these conditions injects a torque

step whose magnitude equals the full gated component; at 100 Hz with a 400 Nm/s rate limit, such a step is spread over several control cycles by the rate limiter and appears to the plant as an unmodelled impulse. Smoothstep removes the step entirely: the gated component enters and leaves the torque sum with continuous first derivative, so the rate limiter is never the active constraint during a gate transition. The one exception is g_{flight} , which is a genuinely discrete load-based hysteresis with asymmetric engage/release timers (2-step engage, 5-step release); contact loss is a binary physical event, and the asymmetric timers exist specifically to prevent balance torque from re-engaging during landing bounce. Note that the composition $g_{\text{env}} = 1 - \text{ss}(\text{EMA}(|v_{\text{sag}}|); \cdot)$ is only C^0 in time, because the asymmetric EMA of Eq. (14) switches coefficients when $|v_t|$ crosses EMA_{t-1} ; the smoothstep is still C^1 in its argument, so the resulting torque is continuous but its time derivative may jump at attack/release transitions.

Wheel torques control sagittal balance and yaw; leg-joint torques control posture and per-leg height adaptation. The two channels are coupled only through the plant, not through the controller: no term in τ_w reads a leg-posture state and no term in τ_q reads the sagittal balance state. This is what allows flight recovery and terrain adaptation to be added as independent extensions (§IV-E, §IV-F) without retuning the anchor.

Relation to existing gated architectures. “Conditionally-gated” here denotes smoothstep activation by *physical* conditions (g_{prox} , g_{env} , g_{θ} , g_{terrain}), and we make no claim that continuous blending is itself novel—split-range, selector, and override controllers have used it in process control for decades [16], [17]. Three distinctions are worth stating precisely. First, industrial override architectures select *among parallel controllers* using process-variable limits; ACC gates *components within a single torque assembly* using physical regime. Second, the gating variables are not control errors but physical quantities of three different kinds—spatial (distance from home), temporal (velocity envelope), and safety-interlock (attitude)—each chosen because a specific measured failure mode of this platform is expressible in it (§VI). Third, ACC is not cascade control [15]: there is no inner/outer loop hierarchy and no setpoint is passed between controllers. The contribution is therefore the *synthesis*—which gating surfaces, on which variables, with which asymmetry—not the blending mechanism.

B. Sagittal Balance Core (τ_{balance})

The sagittal balance component is always active and acts entirely in the wheel-torque channel:

$$\tau_{\text{balance}} = \begin{bmatrix} \tau_{\text{com}} + \tau_{\text{diff}} \\ \tau_{\text{com}} - \tau_{\text{diff}} \end{bmatrix}, \quad \begin{aligned} \tau_{\text{com}} &= \tau_{\text{pitch}} + \tau_{\text{damping}} + \tau_{\text{pos}}, \\ \tau_{\text{diff}} &= \tau_{\text{lat}} + \tau_{\text{yaw}}. \end{aligned} \quad (3)$$

where τ_{com} and τ_{diff} are common-mode and differential components, respectively. The individual terms are:

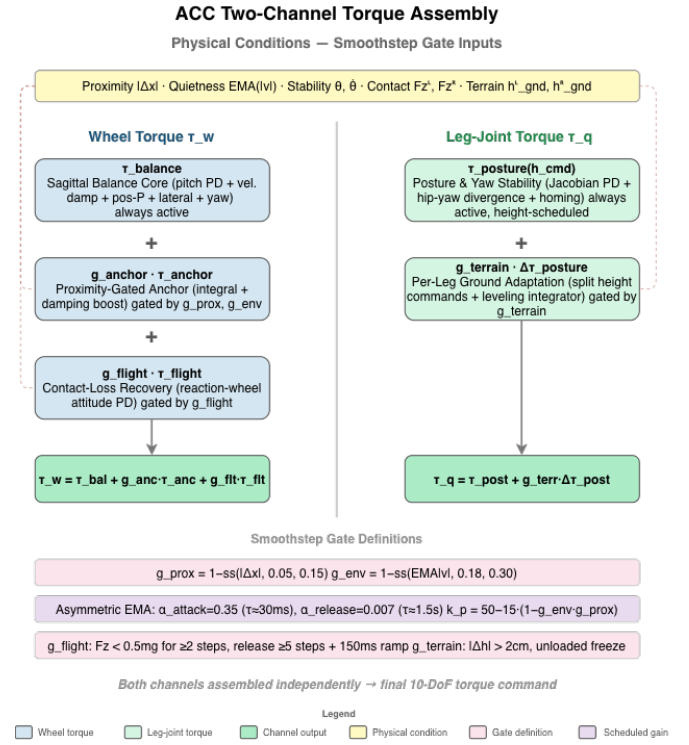


Fig. 2. ACC two-channel torque assembly. Physical conditions gate each component via continuous smoothstep functions (dashed).

$$\tau_{\text{pitch}} = k_p \cdot \theta + k_d \cdot \dot{\theta}, \quad (4)$$

$$\tau_{\text{damping}} = -k_{\text{vel}} \cdot v_{\text{sag}}, \quad (5)$$

$$\tau_{\text{pos}} = -k_{\text{pos}} \cdot \text{clip}(\Delta x, \pm 0.10 \text{ m}), \quad (6)$$

$$\tau_{\text{lat}} = k_{p,\text{roll}} \cdot \phi + k_{d,\text{roll}} \cdot \dot{\phi}, \quad (7)$$

$$\tau_{\text{yaw}} = k_{p,\text{yaw}} \cdot e_{\psi} + k_{d,\text{yaw}} \cdot \omega_z, \quad (8)$$

where θ is the torso pitch angle, $\dot{\theta}$ is the pitch rate, v_{sag} is the sagittal body velocity, Δx is the sagittal position error from the latched home position, ϕ is the roll angle, and e_{ψ} is the yaw error.

P-only limitation. The P position term (k_{pos} , clipped to $\pm 0.10 \text{ m}$, saturating at $\pm 4 \text{ Nm}$) keeps the robot within $\pm 4 - 6 \text{ cm}$ of home but leaves a 1.3 Nm equilibrium bias sustaining a perpetual limit cycle—motivating the anchor integral.

Why saturation-triggered anti-windup is insufficient. Saturation-triggered AW triggers on actuator saturation; on this platform the critical failure is **state-dependent**: during a push, position error grows to 10–30 cm while wheels remain within their 20 rad/s range, so saturation-triggered AW never activates. ACC’s proximity gate provides spatial confinement via continuous smoothstep—avoiding both over-restriction (blocking slow-drift correction) and under-restriction (allowing windup during sustained pushes). The simplest conditional-integration AW is evaluated in Appendix A; broader AW families remain open.

Pitch stiffness. Pitch stiffness is fixed at $k_p=50$ in the canonical ACC architecture. The anchor and boost alone resolve the precision-push trade-off without gain scheduling; a scheduled variant that softened k_p from 50 to 35 during pushes was tested and rejected—it did not improve push survival (S0 vs. S1, Table IV) and is excluded from the canonical specification.

C. Proximity-Gated Anchor Mechanism ($g_{\text{anchor}} \cdot \tau_{\text{anchor}}$)

The anchor mechanism is the core contribution. It adds two terms to the wheel-torque channel that are active only when the robot is near home *and* genuinely at rest:

$$\tau_{\text{anchor}} = K_i \int g_{\text{prox}} \cdot g_{\text{env}} \cdot (-\Delta x) dt + g_{\text{boost}} \cdot K_{\text{boost}} \cdot (-v_{\text{sag}}), \quad (9)$$

where all gates are continuous smoothstep functions designed to equal 1 when the robot is “at home and quiet” and transition smoothly to 0 otherwise:

$$g_{\text{prox}}(\Delta x) = 1 - \text{ss}(|\Delta x|; 0.05 \text{ m}, 0.15 \text{ m}), \quad (10)$$

$$g_{\text{env}}(v_{\text{sag}}) = 1 - \text{ss}(\text{EMA}(|v_{\text{sag}}|); 0.25 \text{ m/s}, 0.50 \text{ m/s}), \quad (11)$$

$$g_{\theta}(\theta, \dot{\theta}) = [1 - \text{ss}(|\theta|; 2^\circ, 12^\circ)] \times [1 - \text{ss}(|\dot{\theta}|; 2^\circ/\text{s}, 15^\circ/\text{s})], \quad (12)$$

$$g_{\text{boost}} = g_{\text{prox}} \cdot g_{\text{env}} \cdot g_{\theta}, \quad (13)$$

where $\text{ss}(x; a, b)$ denotes the smoothstep of Eq. (2) with lower and upper knees a and b . Each gate is therefore a dimensionless, monotone, C^1 function that equals 1 inside its “home and quiet” band, falls smoothly to 0 outside it, and never switches discontinuously. Three physically distinct conditions must hold simultaneously for the anchor to integrate: the robot must be spatially near home (g_{prox}), temporally quiet (g_{env}), and attitudinally settled (g_{θ}).

The pitch stability gate g_{θ} closes when either pitch or pitch rate exceeds its band, preventing the damping boost from fighting a gentle sustained push that does not trigger the velocity envelope. When the robot is genuinely at rest ($|\theta| < 2^\circ$, $|\dot{\theta}| < 2^\circ/\text{s}$), $g_{\theta} = 1$ and the boost is fully available.

Asymmetric envelope follower. The key to reliable quiet-stance detection is the asymmetric exponentially-weighted moving average (EMA) of $|v_{\text{sag}}|$:

$$\text{EMA}_t = \begin{cases} \alpha_a |v_t| + (1 - \alpha_a) \text{EMA}_{t-1}, & \text{if } |v_t| > \text{EMA}_{t-1}, \\ \alpha_r |v_t| + (1 - \alpha_r) \text{EMA}_{t-1}, & \text{otherwise,} \end{cases} \quad (14)$$

with time constants $\tau_{\text{attack}} \approx 23 \text{ ms}$ and $\tau_{\text{release}} \approx 1.5 \text{ s}$ as the primary specification; the discrete coefficients are derived via $\alpha_a = 1 - \exp(-\Delta t / \tau_{\text{attack}}) \approx 0.35$ and $\alpha_r = 1 - \exp(-\Delta t / \tau_{\text{release}}) \approx 0.007$ at 100 Hz ($\Delta t = 0.01 \text{ s}$). This asymmetry is essential: a symmetric EMA is *bistable*—post-push oscillation holds the EMA in a middle band where the gate is half-open and the boost cannot collapse the cycle (validated by the symmetric-EMA ablation, Section V). The fast-attack/slow-release envelope jumps to 1 only when oscillation truly decays,

and drops to 0 within $\sim 25 \text{ ms}$ of a disturbance. The concept traces to peak-hold and envelope-detector circuits [16]; ACC’s innovation is coupling envelope detection to a spatial proximity gate for integral conditioning. Fig. 3 illustrates the complete gate activation sequence during a push-recovery cycle.

EMA initialization. $\text{EMA}_0=0$ biases $g_{\text{env}} \approx 1$ for $\sim 7.5 \text{ s}$ at startup, decaying as velocity noise accumulates. Measurements use 3–5 s settle (see Table captions); hardware should initialize $\text{EMA}_0=1.0$.

D. Posture and Yaw Stability (τ_{posture})

Posture regulation uses Jacobian-based PD gains scheduled across 23 heights:

$$\tau_{\text{posture}} = \tau_{\text{PD}}^{\text{jac}}(q, \dot{q}, h^{\text{cmd}}) + \tau_{\text{div}}(q_{\text{hy}}^L, q_{\text{hy}}^R) + g_{\text{settled}} \tau_{\text{homing}}(q - q_{\text{ref}}), \quad (15)$$

where τ_{div} damps hip-yaw divergence and τ_{homing} restores nominal posture when upright and settled. Yaw return uses wheel-differential torque.

E. Contact-Loss Recovery ($g_{\text{flight}} \cdot \tau_{\text{flight}}$)

When both wheels lose ground contact, ACC detects this via per-wheel normal forces and substitutes a flight attitude controller using the wheels as reaction flywheels:

$$g_{\text{flight}} = \begin{cases} 1, & \text{if } F_z^L, F_z^R < 0.5mg \text{ for } \geq 2 \text{ steps,} \\ 0, & \text{if } \min(F_z^L, F_z^R) \geq 0.5mg \text{ for } \geq 5 \text{ steps,} \end{cases} \quad (16)$$

$$\tau_{\text{flight}} = \text{clip}(2.0\theta + 0.5\dot{\theta} - 0.02\omega_{\text{wheel}}, \pm 2 \text{ Nm}). \quad (17)$$

Forward wheel spin pitches nose-up during descent; 5-step release hysteresis prevents balance-torque re-engagement during bounce. The free-fall drop is a stress test and the ledge drive-off the intended use case. We note in advance of the results that this override is the one ACC mechanism whose ablation does *not* show it to be necessary at the reported scales: the wheel-inverted-pendulum loop it replaces is itself a reaction-flywheel controller when the wheels are unloaded, and the override’s measured contribution is to bound landing-attitude variance near the upper end of the drop envelope rather than to enable recovery (§V-C, Table V).

F. Per-Leg Ground Adaptation (g_{terrain})

When the two wheels rest at different ground heights—a curb straddle, an oblique ledge—a single torso height command forces the torso to roll by the step angle. ACC instead maintains an *independent ground estimate per leg* and issues two height commands. The estimate is taken from the wheel contact-point height, which is exact at any body tilt for a rigid wheel, and is updated only while that wheel carries real load:

$$\hat{h}_{\text{gnd}}^i \leftarrow \hat{h}_{\text{gnd}}^i + \alpha_g (z_{\text{contact}}^i - \hat{h}_{\text{gnd}}^i), \quad F_z^i \geq 0.2mg, \quad i \in \{L, R\}, \quad (18)$$

with $\alpha_g = 0.35$ per control step to reject contact chatter. The $0.2mg$ load test rather than mere geometric touching is deliberate: a lightly touching wheel that is in the act of

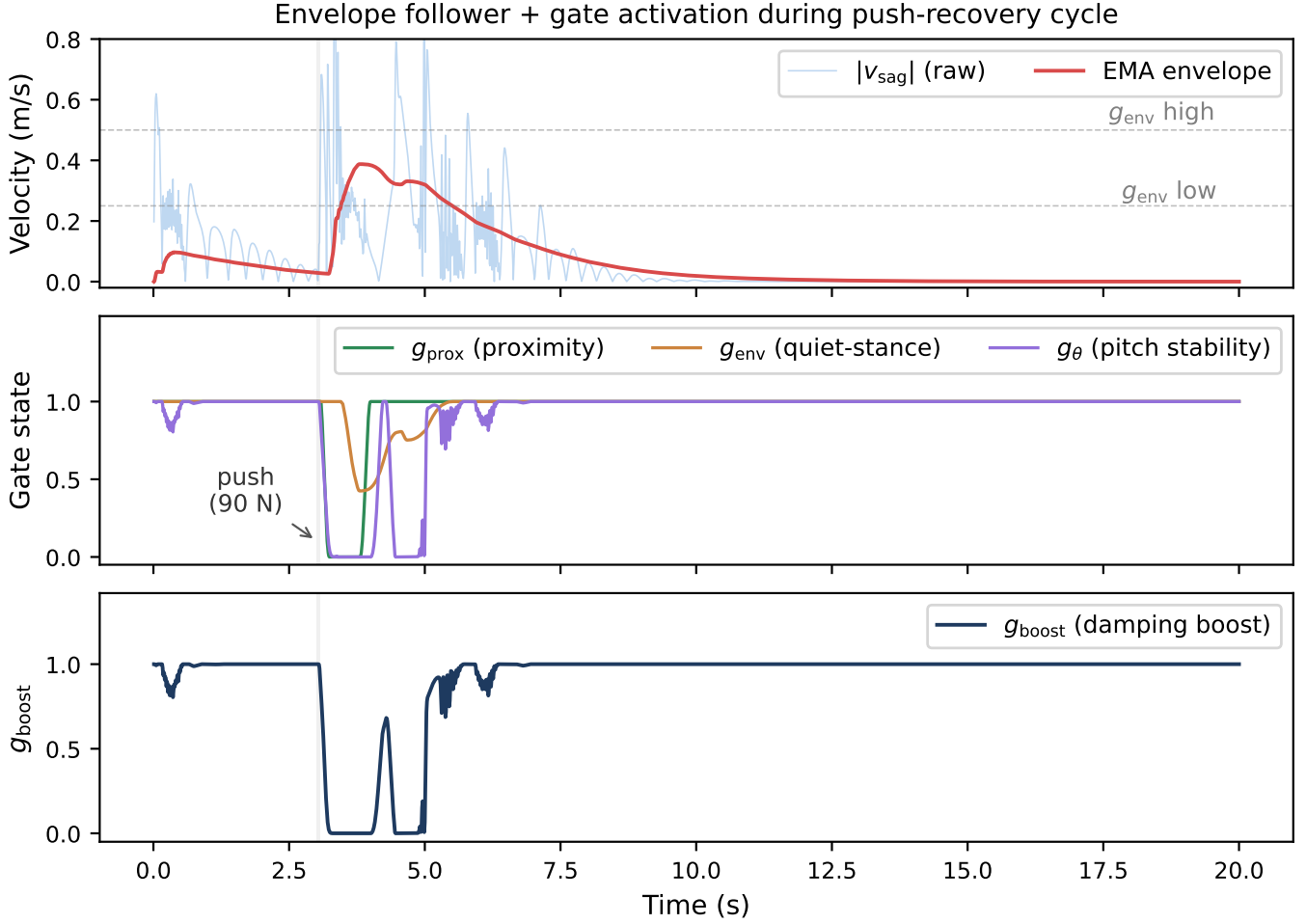


Fig. 3. Gate activation during a 90 N lateral (world +x) push-recovery cycle (100Hz control). **Top:** $|v_{\text{sag}}|$ and asymmetric EMA envelope; dashed lines mark the 0.25–0.50 m/s quiet-stance band. **Middle:** Individual gate states. **Bottom:** Composite damping boost g_{boost} . Grey shading: 7-step push window. The three panels are the empirical basis for finding (iii) of §V-A4: g_{prox} and g_{env} both sit pinned at their quiet-stance values before the impulse and after the ringdown, which is why ablating either reproduces ACC bit-for-bit at idle, and why only a disturbance experiment can measure their contribution.

unloading rides a few centimetres high and injected a 2.9 cm phantom ground step on the 15 cm curb.

An unloaded wheel does *not* update (18). It freezes its estimate, because an airborne wheel supplies no information about where its ground is. The one exception is the only physically unambiguous “my ground is gone” signal—the wheel centre descending *below* its own believed ground:

$$\delta^i > 3 \text{ mm} \Rightarrow \hat{h}_{\text{gnd}}^i = -(v_0 + k_s \delta^i), \quad (19)$$

where $\delta^i = \hat{h}_{\text{gnd}}^i - z_{\text{contact}}^i$ is the depth by which the wheel has fallen below its believed ground, with $v_0 = 0.5 \text{ m/s}$ and $k_s = 50 \text{ s}^{-1}$. The proportional term matters: a fixed slew converged the ground estimate in 0.30 s, by which time a 20 cm dismount had already tipped the robot past recovery, whereas a 15 cm drop under (19) slews at 8 m/s and converges within two control steps, so the leg extends *during* the fall rather than after it. When both wheels are airborne, both estimates track the mean wheel height, which reproduces the

symmetric pre-adaptor behaviour exactly and leaves ballistic flight to §IV-E.

The step $\Delta h = \hat{h}_{\text{gnd}}^L - \hat{h}_{\text{gnd}}^R$ is then absorbed by the legs rather than the torso. The split is kinematic, not a linear height offset: with $L(\cdot)$ the forward-kinematic leg length and L^{-1} its inverse over the calibrated posture range, the leg on the higher ground is shortened by the full step,

$$L_{\text{high}} = L(h^{\text{cmd}}) - |\Delta h|, \quad (20)$$

$$h^{\text{high}} = L^{-1}(L_{\text{high}}), \quad h^{\text{low}} = h^{\text{cmd}}, \quad (21)$$

both clamped to the calibrated envelope extended by 10 cm at each end. To first order this is the familiar $h^{\text{cmd}} \mp \Delta h/2$ symmetric split, but the $h \mapsto L$ map is markedly nonlinear near full squat: the linear proxy erred by 12 cm at the lower extrapolation limit, folding the knee back under the thigh. If L_{high} falls below the shortest physically reachable leg, ACC raises the *whole* robot until the short leg fits, trading commanded height for the ability to span the step. A 20 cm curb at nominal stance exceeds even this: it would require 170°

TABLE I

CAPABILITY COMPARISON ACROSS WHEELED BIPED PLATFORMS. “—” = NOT REPORTED IN CITED WORK. IDLE PRECISION AND PUSH FORCE ARE NOT STANDARDIZED BENCHMARKS; QUANTITATIVE CROSS-PLATFORM COMPARISON IS LIMITED BY METRIC HETEROGENEITY.

Metric	Ascento	DIABLO	Whaleper	SKATER	ACC (Ours)
Balance	LQR	Cascaded LQR	LQR + PPO	Convex MPC	Gated PD + anchor I
Idle precision	—	—	—	—	0.294 mm sag. CoM RMS
Push recovery	— ^a	—	—	—	8-dir., $F_{\text{med}}=117$ N
Height	Discrete jump	Stance ^b	Multi-modal	Contin.	0.35–0.45 m CoM^c
Flight	Jumping (airtime)	—	—	—	Drop 100 cm, ledge 50 cm
Terrain	—	—	—	Vision-based	Per-leg contact
Validation	Hardware	Hardware	Hardware	Hardware	Simulation

^aQual. omnidirectional push recovery demonstrated in supplementary video; quantitative force thresholds not reported in cited work. ^bHeight variation demonstrated; quantitative range not reported.

^cCalibrated CoM-z band (§III-A), ± 5 cm about the 0.404 m nominal, with joint targets extrapolated to 0.254–0.554 m; commanded ± 5 cm transitions are evaluated under disturbance in §V-F. Not comparable to the base-z ranges of the cited platforms or of our PPO baseline.

of knee flexion against a 2.7 rad limit, and the uncompensated remainder appears as a predicted torso roll $\arctan(-\varepsilon/w_{\text{track}})$ with ε the unspanned step and $w_{\text{track}} = 0.278$ m the wheel separation. Reporting this expected roll explicitly is what keeps a legitimate straddle from being misread as an incipient fall by the tilt-based safety monitors.

Because the $h \mapsto L$ table is itself extrapolated at the extremes, a purely feedforward split left a systematic 6° torso roll on the 10 cm curb. A closed-loop leveling integrator trims the split by the *measured* roll at 0.15 m/(rad s) with ± 6 cm of authority. It integrates only when all three of the following hold: both wheels loaded, a genuine ground step $|\Delta h| \geq 2$ cm, and a ground estimate settling at less than 0.15 m/s. The gating is not cosmetic. Without the step condition, the dynamic roll of a turn is integrated as if it were terrain and wound an 8.6° roll bias through a 180° turn; without the rate condition, the geometric transient of a curb climb is integrated as a leveling error and compresses the flat leg. Off-step, the trim decays to zero with a 1 s time constant.

G. Complete Numerical Parameter Set

Table II lists core parameters; the full set (50+ gains) is available in the supplementary material.

V. EXPERIMENTAL VALIDATION

We evaluate ACC across four scenario classes, each mapped to the component it tests and paired with targeted ablations. ACC is evaluated at 500 Hz simulation / 100 Hz control; the PPO and LQR baselines run at 50 Hz, their own design rate. All configurations use a 400 Nm/s torque rate limit per joint.

TABLE II

CORE NUMERICAL PARAMETERS FOR THE ACC BALANCE CONTROLLER. ALL VALUES IN SI UNITS UNLESS NOTED OTHERWISE.

Balance Core		
k_p	50	Nm/rad
k_d	3.0	Nm/(rad/s)
k_{pos}	20	Nm/m
Position clip	± 4.0	Nm
k_{vel}	15	Nm/(m/s)
Vel. damp. scale	1.5	—
k_{roll}	55	Nm/rad
$k_{\text{roll},d}$	5.5	Nm/(rad/s)
k_{yaw}	1.0	Nm/rad
$k_{\text{yaw},d}$	0.3	Nm/(rad/s)
$k_{\text{pos},\text{return}}$	2.0	Nm/m
k_{heading}	2.0	Nm/rad
$k_{\text{heading},d}$	0.6	Nm/(rad/s)
Anchor Mechanism		
K_i (anchor integral)	4.0	Nm/(m·s)
Integral cap	2.0	Nm
Integral leak	5×10^{-3}	per step
Boost scale k_{vb}	5.0	—
g_{prox} band	0.05–0.15	m
g_{env} band	0.25–0.50	m/s
τ_{attack}	0.023	s
τ_{release}	1.5	s
$g_{\text{stab}}(\theta)$	2–12	$^\circ$
$\dot{g}_{\text{stab}}(\dot{\theta})$	2–15	$^\circ/\text{s}$
Flight / Terrain / Other		
k_{flight} (P / D)	2.0 / 0.5	—
Wheel damping (flight)	0.02	Nm/(rad/s)
Flight engage / release	2 / 5	steps
F_z thresh. (flight / terrain)	0.5 / 0.2	$\times mg$

Control-rate confound. ACC therefore enjoys a $2\times$ control-bandwidth advantage over the baselines, and we were not able to remove it. Re-running ACC at 50 Hz with the discrete coefficients recomputed for the longer step ($\alpha_a \approx 0.58$ and $\alpha_r \approx 0.013$, re-derived from Eq. (14) at $\Delta t = 0.02$ s, with the anchor leak rescaled to 0.010 per step) does not preserve the controller: the robot collapses, and the mean CoM height over the recorded window is 0.124 m against a nominal standing height of 0.404 m. The 0.050 mm “idle RMS” and the 10 N push threshold produced by that run are artifacts of a robot lying motionless on the ground—10 N is the lower bound of the binary search, returned when even the smallest tested impulse fails. We consequently report *no* 50 Hz ACC result and make no claim that the architecture is rate-independent. What Table VII establishes is that the classical baselines fail at the rate for which they were designed; it does not establish that they would fail at 100 Hz, nor that ACC would succeed at 50 Hz. Restoring rate parity requires re-tuning the ACC gains for the 20 ms step—a step-response and gain-margin re-derivation we have not performed—and is stated as an open limitation (Limitation 5) rather than a resolved confound. RNG seeds are deterministic: idle standing uses seed = 20260727 + trial_id; push ablation uses seed = 20260728 $\times$ 100 + rep; LQR baselines cycle seeds {0, 42, 123} across episodes.

A. Anchored Standing

1) Two precision metrics, deliberately kept separate:

“Standing precision” names two different quantities in the wheeled-biped literature, and they are not interchangeable. Let $\mathbf{p}(t) \in \mathbb{R}^2$ be the horizontal base position over an evaluation window of length T , let $\mathbf{p}_0$ be the commanded home position, and let $\bar{\mathbf{p}} = \frac{1}{T} \int_0^T \mathbf{p}(t) dt$ be the trial mean. We report

$$A = \left(\frac{1}{T} \int_0^T \|\mathbf{p}(t) - \mathbf{p}_0\|^2 dt \right)^{1/2}, \quad (22)$$

$$R = \left(\frac{1}{T} \int_0^T \|\mathbf{p}(t) - \bar{\mathbf{p}}\|^2 dt \right)^{1/2}. \quad (23)$$

A is the *accuracy*: RMS displacement from the position the controller was told to hold. It retains any DC parking offset, so it answers “how close to the commanded pose does the robot actually park?”. R is the *repeatability*: RMS about the trial’s own mean, with the DC offset removed. It answers “how still is the robot once it has parked?”. A controller with a persistent 27 mm offset and a perfectly steady hold scores $A=27$ mm but $R \approx 0$ mm; a controller centred on the target but oscillating scores the reverse. Reporting either alone is uninformative, and—because the two differ by more than an order of magnitude on this platform—comparing a value of one against a value of the other produces a meaningless ratio.

Both are reported for every configuration below, resolved separately along the sagittal axis, the lateral axis, and the full plane, because the two horizontal axes of this platform are not merely different in degree—they are actuated by different means and must not be pooled or interchanged. We therefore fix the convention explicitly, since the two are easy to transpose and the physical conclusions invert if they are:

- The **sagittal** axis is world y . Both wheel axes point along world x and the two wheels are separated along world x ($w_{\text{track}}=0.278$ m), so the rolling direction—the axis on which wheel torque produces ground force, and the only axis with a genuine inverted-pendulum mode—is world y . The controller agrees: `init_v3_controller` packs $\hat{\mathbf{s}} = (0, 1)$ as its sagittal unit vector.
- The **lateral** axis is world x . No actuator produces lateral ground force; the axis is held passively by the wheel contacts and by the roll restoring moment of the track width.

Two dynamic checks confirm the assignment rather than merely asserting it. A 90 N impulse along y is absorbed with a peak wheel speed of 52 rad/s and leaves a residual offset of 4.8 mm—the wheels drive back under the CoM—whereas the same impulse along x leaves a permanent 23.2 mm offset at a peak wheel speed of 18 rad/s, because nothing can push the robot sideways again. And over a 20 s quiet stance the robot rolls 24.6 mm along y with the wheel midpoint tracking the CoM to within 0.2 mm, while x holds to 0.02 mm. All four checks are reproduced by `scripts/verify_body_axes.py`, which asserts on disagreement.

This matters for reading Table III: the anchor integral and the gated damping boost both live in the wheel-torque channel, so *both act on the sagittal axis only*. The lateral column is therefore a measure of how little the plant moves when nothing drives it, not a measure of controller quality, and the sagittal and planar columns carry essentially all of the information about what ACC does (Sec. VI, Limitation 8).

2) *Unified measurement protocol*: Every row of Table III was re-measured under one protocol so that no two entries in the same column come from different definitions. Each trial runs 25 s at 500 Hz physics / 100 Hz control from the nominal standing pose with randomized initial conditions ($\sigma=0.005$ rad per joint, clipped to joint limits; $\sigma=1$ mm on root height), seeded `seed=20260727+trial_id` for `trial_id` $\in [0, 10)$. The first 5 s are discarded as settling—long enough to cover the $\text{EMA}_0=0$ transient (~ 7.5 s to full release, but < 5 s to within 1 % of the quiet-stance value)—and A and R are computed over the remaining 20 s. Displacements are taken relative to each trial’s own $t=0$ pose, so $\mathbf{p}_0$ is the commanded home rather than the world origin. A trial is scored as a fall if $|\theta| > 46^\circ$ or root height < 0.15 m at any point. Confidence intervals are Student- t ($t_{9,0.025}=2.262$, $\text{SEM}=\text{SD}/\sqrt{10}$).

We note explicitly what this protocol *cannot* test: because every trial starts at the commanded home, it never exercises the anchor’s return-to-home function. It measures the controller’s ability to *stay* put, not to *come back*. The latter is measured by the push experiments of Sec. V-B. This distinction turns out to matter for attributing the mechanism (Sec. V-A4).

3) *Result*: ACC holds **sub-millimeter sagittal repeatability**, $R_{\text{sag}} = 0.294 \pm 0.015$ mm (95% CI $[0.283, 0.305]$ mm), across ten randomized trials with no falls. This is the figure of merit for the platform, because sagittal is the actuated inverted-pendulum axis: it is the axis on which the robot can fall, the axis the wheel-torque channel drives, and the axis that every configuration in the ladder is trying to hold.

We deliberately do not quote a signal-to-noise ratio against a simulator noise floor here. The floor is measured by locking the joints and pinning the base (`run_r2_between_trial_stats.py--noise-floor`), and that procedure only zeroes velocities rather than welding the base, so it is trustworthy on the laterally-constrained axis (0.0019 mm RMS) but not on the gravity-loaded sagittal and vertical axes, where the residual sag it reports (5.79 mm and 63.5 mm) is an artifact of the pinning rather than integrator noise. The relevant scale for R_{sag} is instead the spread *between* configurations measured under the identical protocol: removing a single mechanism (S2) moves R_{sag} from 0.294 to 4.772 mm, a factor of 16 with non-overlapping SDs, which is what establishes that the metric resolves the effect being studied.

Against the P-only baseline (L0), which sustains the 69.7 mm peak-to-peak sagittal limit cycle described in Sec. I, the like-for-like improvements are

- sagittal repeatability R_{sag} : $17.461 \rightarrow 0.294$ mm, a factor of **59.4**;
- planar repeatability R_{xy} : $17.845 \rightarrow 0.788$ mm, a factor of **22.6**;

TABLE III

IDLE STANDING PRECISION ACROSS THE FULL ABLATION LADDER. ALL ROWS RE-MEASURED UNDER THE SINGLE PROTOCOL OF SEC. V-A; VALUES ARE MEAN $\pm$ SD OVER $N=10$ TRIALS. SEE THE NOTE BELOW THE TABLE FOR COLUMN DEFINITIONS.

Configuration		R_{sag} (mm) sagittal rep.	R_{xy} (mm) planar rep.	R_{lat} (mm) lateral rep.	$\bar{s}$ (mm) sag. offset	A_{lat} (mm) lateral acc.	Survival
Additive ladder (bottom-up)							
L0	P-only (PD, no position, no integral)	17.461 ± 0.258	17.845 ± 0.279	3.675 ± 0.246	-44.8	9.933 ± 0.712	10/10
L1	+ position homing (± 0.10 m)	20.397 ± 0.017	20.414 ± 0.019	0.834 ± 0.066	-30.4	2.629 ± 0.884	10/10
L2	+ ungated integral K_i	25.026 ± 0.039	25.056 ± 0.043	1.212 ± 0.110	-27.5	3.661 ± 1.264	10/10
L2.5	+ proximity gate only (no envelope, no boost)	25.198 ± 0.017	25.227 ± 0.020	1.205 ± 0.099	-27.6	4.023 ± 1.194	10/10
Proposed controller							
L3/S1	ACC (full anchor, fixed $k_p=50$)	0.294 ± 0.015	0.788 ± 0.069	0.730 ± 0.073	-27.0	1.213 ± 0.554	10/10
Subtractive ablation (top-down from ACC)							
S0	ACC + scheduled k_p (rejected variant)	0.294 ± 0.015	0.788 ± 0.069	0.730 ± 0.073	-27.0	1.213 ± 0.554	10/10
S2	- gated damping boost	4.772 ± 0.069	4.824 ± 0.067	0.703 ± 0.069	-26.8	1.260 ± 0.616	10/10
S3	- asymmetric envelope (symmetric EMA)	0.294 ± 0.015	0.788 ± 0.069	0.730 ± 0.073	-27.0	1.213 ± 0.554	10/10
S4	- proximity gate (integral always on)	0.294 ± 0.015	0.788 ± 0.069	0.730 ± 0.073	-27.0	1.213 ± 0.554	10/10
S5	- pitch safety gate g_θ	—	—	—	—	—	0/10
Single-mechanism isolation (ACC with one term disabled)							
M1	$K_i=0$ (anchor integral removed, gates intact)	0.089 ± 0.035	0.733 ± 0.072	0.727 ± 0.071	-31.2	1.192 ± 0.535	10/10
M2	damping terms reverted to their L2 values	25.198 ± 0.017	25.227 ± 0.020	1.205 ± 0.099	-27.6	4.023 ± 1.194	10/10
M3	$K_i=0$ and boost = 0	3.973 ± 0.130	4.037 ± 0.128	0.710 ± 0.069	-31.1	1.241 ± 0.591	10/10

Protocol. $N=10$ trials, 100Hz control, 20s measurement window after a 5s settle, clean sensors, 0ms actuator delay.

Columns. R = repeatability (RMS about the trial mean, Eq. (23)); A = accuracy (RMS from commanded home, Eq. (22)); $\bar{s}$ = mean signed sagittal parking offset. Sagittal is world y and lateral is world x (Sec. V-A). The anchor integral and the damping boost act on the sagittal axis only, so R_{sag} and $\bar{s}$ are the columns that discriminate between configurations, while R_{lat} measures a passively held axis and is near-constant by construction.

Omitted columns. A_{sag} and A_{xy} carry no information beyond the columns shown: over all twelve surviving rows the identities $A_{\text{sag}} = \sqrt{\bar{s}^2 + R_{\text{sag}}^2}$ and $A_{xy} = \sqrt{A_{\text{sag}}^2 + A_{\text{lat}}^2}$ hold to 0.003 mm and 0.02 mm respectively—accuracy on an axis is just that axis’s parking offset and its repeatability added in quadrature. Produced by `scripts/collect_idle_ladder.py` (per-trial worker `scripts/_idle_ladder_worker.py`); raw output `outputs/paper_verification/idle_ladder.json`.

- lateral repeatability R_{lat} : $3.675 \rightarrow 0.730$ mm, a factor of **5.0**.

The ordering of these three factors is itself informative and worth stating plainly, because it is the opposite of what a reader might expect. The largest gain is on the sagittal axis, which is the hard axis; the smallest is on the lateral axis, which is the easy one. That is the correct signature for a wheel-torque controller: every term ACC adds beyond L0—position homing, the anchor integral, the gated damping boost—produces ground force along the rolling direction and none of them can produce lateral force at all. The residual $5\times$ lateral improvement is an indirect consequence of the robot pitching and yawing less, not of any lateral control action, and we do not claim it as one. A reader comparing against platforms that report a single scalar “standing precision” should compare against R_{sag} , not against R_{lat} or a planar norm, since the latter two are diluted by an axis this class of platform does not actuate.

The accuracy picture is markedly worse than the repeatability picture, and on the same axis. ACC parks $\bar{s} = -27.0$ mm from the commanded home along the sagittal axis and holds that offset to within 0.009 mm SD across trials—the robot rolls backwards by 27 mm during the settle and then stands there, exceptionally still, in the wrong place. Planar accuracy is therefore $A_{xy} = 26.994$ mm and improves only $1.8\times$ over L0 ($49.1 \rightarrow 27.0$ mm), a full order of magnitude worse than the repeatability gain. The offset is not a measurement artifact and

not an unregulated axis: it lies precisely on the axis the anchor integral does control, and the integral reduces it only from 31.2 mm (M1, $K_i=0$) to 27.0 mm. An ideal integrator would drive it to zero. We report the discrepancy as the principal open weakness of the present design (Limitation 8) rather than absorbing it into a planar average where it is less visible.

4) *Which mechanism produces the precision:* The bottom two blocks of Table III isolate the contributions. Four findings are worth stating precisely, because three of them contradict the attribution that the gate structure invites.

(i) *Idle steadiness comes from the damping ladder; the integral buys accuracy instead.* The two contributions separate cleanly once the axes are kept apart, and they do not compete for the same metric.

M1 disables the anchor integral entirely ($K_i=0$) while leaving every gate in place, and its *repeatability* is not merely as good as ACC’s but slightly better (R_{sag} 0.089 mm vs. ACC’s 0.294 mm; R_{xy} 0.733 mm vs. 0.788 mm). Its *accuracy*, however, is worse by 4.2 mm: it parks at $\bar{s} = -31.2$ mm against ACC’s -27.0 mm. That is the integral’s entire measurable idle signature, and it is exactly the trade one expects from an integral term—it walks the robot closer to the commanded home at the cost of a small amount of steadiness, since the accumulating term is itself a slowly-varying command. A protocol that reports repeatability alone will conclude, wrongly, that the anchor integral is inert at idle; a protocol that reports accuracy alone will miss that it costs a factor of three in

steadiness.

Conversely M2 keeps the integral and every gate but reverts the two velocity-damping coefficients to their L2 values, and reproduces the L2/L2.5 limit cycle exactly ($R_{\text{sag}} = 25.198 \text{ mm}$). Steadiness is therefore attributable to the damping terms, in two stages: raising the velocity-damping scale takes R_{sag} from 25.2 mm to 3.97 mm (M3, boost still off), and enabling the gated damping boost takes it from 3.97 mm to 0.294 mm (ACC)—a combined factor of 86 decomposed into factors of ~ 6.3 and ~ 13.5 . S2 confirms the second stage independently from the top down: removing only the boost degrades R_{sag} from 0.294 mm to 4.772 mm.

(ii) *The damping boost acts sagittally, as it must.* S2 degrades sagittal repeatability by a factor of 16 ($0.294 \rightarrow 4.772 \text{ mm}$) while leaving lateral repeatability statistically unchanged ($R_{\text{lat}} = 0.703 \text{ mm}$ vs. ACC's 0.730 mm , overlapping CIs). This is the only outcome consistent with the architecture: the boost multiplies a velocity-damping coefficient inside the wheel-torque channel (Eq. (3)), and wheel torque generates ground force along the rolling direction only. The residual oscillation it suppresses is therefore a sagittal pitching/rolling-forward mode, and an evaluation that reported the lateral axis alone would conclude, wrongly, that the boost does nothing at idle. We flag this explicitly because the lateral column is the one that looks like a plausible “x-axis RMS” to report by default, and it is the one column in Table III that is nearly blind to every mechanism in the ladder: across all twelve surviving configurations R_{lat} spans only $0.70\text{--}3.68 \text{ mm}$, and across the seven rows that retain the full damping ladder (S0–S4, M1, M3) only $0.70\text{--}0.73 \text{ mm}$.

(iii) *The proximity gate and the asymmetric envelope are inert during quiet stance.* S4 (proximity gate removed) and S3 (symmetric envelope) are *bit-identical* to ACC on every idle metric. This is the expected behavior rather than a null result: during quiet stance the base never leaves the 5 cm inner band, so $g_{\text{prox}} \equiv 1$ and gating it changes nothing; and the velocity envelope never rises into the attack branch, so the asymmetric and symmetric followers produce the same signal. Both mechanisms are defined by what they do when the robot is *displaced*—windup prevention during a push (S4) and ringdown after one (S3)—and are correctly evaluated only in Table IV. The idle protocol, which starts at home and stays there, is blind to both by construction.

(iv) *The pitch safety gate is required even to stand.* S5 falls in 10/10 trials during pure quiet stance, with no push applied. Removing g_{θ} leaves the damping boost free to act on the pitch state at large excursions, and the resulting positive feedback diverges from the $\pm 0.005 \text{ rad}$ initial perturbation alone. This is a stronger safety result than the push-only evidence previously reported for this ablation.

Taken together: the anchor's *gates* are what make an aggressive damping and integral configuration admissible without windup or divergence, while the damping terms they admit are what deliver the steadiness. The gates are enabling conditions, not the proximate cause of the precision—a distinction the earlier framing of this work conflated. The integral's primary

role is drift rejection under disturbance, which by construction cannot appear in a protocol that starts at the target; its measurable idle signature is confined to the 4.2 mm reduction in sagittal parking offset noted above (M1 -31.2 vs. ACC -27.0 mm), bought at a $3\times$ cost in sagittal repeatability. Both effects sit on the sagittal axis, and neither is visible on the lateral one.

Post-push velocity ringdown decays within 9 s under ACC, whereas P-only oscillates indefinitely because the 1.3 Nm equilibrium bias sustains the limit cycle. The symmetric-EMA ablation exhibits bistable failure: a post-push cycle holds the envelope at ≈ 0.10 , leaving the boost gate half-open, and the robot oscillates perpetually. The ungated-integral ablation winds up during the push and prevents recovery entirely. All three of these are post-push behaviors and are quantified in Sec. V-B.

B. Omnidirectional Push Recovery

Push recovery is evaluated via polar sweep: $10\text{--}160 \text{ N}$ impulses (7 control steps) at the torso in 8 directions. Bearings are applied in the world frame as $\mathbf{f} = F(\cos \theta, \sin \theta, 0)$, so $\theta = \pm 90^\circ$ are the sagittal (fore/aft) bearings and $\theta = 0^\circ/180^\circ$ the lateral ones, per the axis convention of Sec. V-A. F_{min} is the maximum magnitude survived in *every* direction; F_{med} is the median. All thresholds measured via binary search over $[10 \text{ N}, 160 \text{ N}]$ (8 iterations per direction, 5 N tolerance). Survival: torso pitch $< 46^\circ$ for 17 s post-push.

Single-bearing stress tests. Several auxiliary experiments in this paper—the robustness sweep (Table VIII), the rate-limit sweep, the ringdown and gate-activation traces (Figs. 3, 5), and the compound-disturbance protocol (Table IX)—apply a single impulse along world $+x$ rather than repeating the full 8-bearing sweep, which would multiply their cost eightfold. That bearing is *lateral*, and it is a below-median one: the polar sweep places 0° at 94.3 N, third-weakest of the eight and 22.6 N below F_{med} . Those experiments are therefore conservative probes of ACC's disturbance margin, not optimistic ones, and we label them as lateral throughout rather than as “forward.” The omnidirectional figures (F_{min} , F_{med}) are the ones that characterize the full envelope.

Table IV reports the factorial ablation. Additively: L1 (homing) adds 4.8 N over L0 ($p < 0.001$); L2 (global integral) adds only 1.3 N over L1 (n.s.); L3 (full anchor) adds 6.7 N over L2 ($p < 0.001$). The L2→L3 step bundles four mechanisms at once, and the push metrics alone cannot separate them; Sec. V-A4 performs that separation on the idle side. What the push data do show is that the proximity gate contributes nothing to push *margin* on its own (L2→L2.5, -0.6 N , n.s.), consistent with its role being windup prevention rather than capture-basin enlargement, and that the full anchor recovers a margin advantage over the ungated integral ($+6.7 \text{ N}$) while simultaneously being the only configuration in the ladder that settles. Subtractively: S1 (fixed $k_p = 50$, canonical) matches or exceeds S0 in all metrics, confirming gain scheduling is unnecessary—the anchor and boost alone resolve the trade-off. The gated damping boost (S2) is the dominant idle-precision

TABLE IV

FACTORIAL ABLATION OF ACC COMPONENTS: PUSH-RECOVERY AND RINGDOWN METRICS. CLEAN SENSORS, ALL ROWS $N=10$, MEAN $\pm$ STD. SEE THE NOTE BELOW THE TABLE FOR DEFINITIONS AND SIGNIFICANCE TESTS.

Configuration	$F_{\min}$ (N)	F_{med} (N)	Ringdown
Additive (bottom-up)			
L0 P-only (PD, no pos., no I)	70.0 $\pm$ 1.0	122.1 $\pm$ 2.1	never
L1 + Pos. homing (± 0.10 m)	74.8 $\pm$ 0.8	130.5 $\pm$ 1.2	never
L2 + Integral K_i (no gate)	76.1 $\pm$ 1.1	132.0 $\pm$ 2.0	—
L2.5 + Prox. gate only (no EMA, no boost)	75.5 $\pm$ 0.9	132.0 $\pm$ 2.3	—
L3 + Full anchor (gate, EMA, boost) ⁷	82.8$\pm$1.0	116.8 $\pm$ 4.7	9.0 s
Subtractive (top-down from ACC)			
S0 ACC + scheduled k_p (rejected) ⁶	79.4 $\pm$ 2.6	114.0 $\pm$ 5.3	9.0 s
S1 ACC ⁶ (canonical, fixed $k_p=50$)	82.8$\pm$1.0	116.8 $\pm$ 4.7	9.0 s
S2 -Damping boost	84.2 $\pm$ 1.1	115.5 $\pm$ 5.3	never ¹
S3 -Asym. EMA (sym. EMA)	81.6 $\pm$ 2.3	109.2 $\pm$ 5.4	never ²
S4 -Prox. gate (global I)	80.0 $\pm$ 2.5	114.0 $\pm$ 5.3	windup ³
S5 -Pitch gate g_θ	8.8 $\pm$ 0.6 ⁵	35.4 $\pm$ 10.3	never ⁴

$F_{\min}$ and F_{med} are the minimum and median survivable push over 8 bearings, each located by binary search to 5 N tolerance. Idle-precision metrics for these same configurations are reported separately in Table III under a single measurement protocol; they are deliberately not duplicated here, because idle and push behaviour are governed by different subsets of the gates.

¹Boost necessary for ringdown. ²Symmetric EMA bistable: post-push oscillation holds EMA in mid-band. ³Without the proximity gate the integral accumulates throughout the displacement and saturates its cap; the robot survives the impulse but does not settle. At idle this ablation is bit-identical to ACC (Table III), since the base never leaves the gate's inner band. ⁴Removing the pitch gate lets the damping boost act on the pitch state at large excursions; the resulting parametric amplification diverges. This ablation additionally fails 10/10 at pure idle with no push applied (Table III). ⁵Below measurement floor (10 N); robot cannot survive minimal push. ⁶S1 (fixed $k_p=50$) is the canonical ACC configuration. S0 (scheduled k_p , softened from 50 to 35 during pushes) was tested and rejected—it did not improve push survival. ⁷L2 $\rightarrow$ L3 = canonical ACC (S1, fixed $k_p=50$); bundles four mechanisms (prox. gate, asym. EMA, boost, pitch gate). S2–S5 isolates each subtractively. Welch's unpaired t -test (two-sided, $N=10$ per group, independent trials; Welch–Satterthwaite $df \in [9, 18]$; uncorrected p , Bonferroni-adj. $\alpha=0.007$, see §VI): L0 $\rightarrow$ L1 +4.8 N ($p<0.001$); L1 $\rightarrow$ L2 +1.3 N (n.s.); L2 $\rightarrow$ L2.5 –0.6 N (n.s., gate alone adds nothing); L2.5 $\rightarrow$ L3 +7.3 N ($p<0.001$). S5 (pitch gate) is critical to safety ($F_{\min}=8.8$ N, and 0/10 survival even at idle); S2 and S3 are required for ringdown. Idle metrics for every row are in Table III. With $N=10$, 95%CI half-width $\approx 0.72 \times \text{std}$ ($t_{0.025,9}=2.262$, single-group $df=9$); sub-2 N differences (e.g., S1 vs. S3 $\Delta F_{\min}=1.2$ N, $CI_{95}(S1) \approx \pm 0.7$ N, $CI_{95}(S3) \approx \pm 1.6$ N) remain non-significant. Produced by scripts/replicate_ablation_n10.py. Raw outputs: rows L2 and S0–S5 in outputs/paper_statistics/ablation_n10_results.json; row L2.5 in outputs/paper_statistics/ablation_n10_L2_5.json; rows L0 and L1 in outputs/paper_statistics/push_ci_all_ablations.json. All at $N=10$ under the same protocol.

mechanism, and it acts sagittally: removing it degrades sagittal repeatability sixteen-fold (0.294 mm $\rightarrow$ 4.772 mm) while leaving lateral repeatability statistically unchanged (0.703 mm vs. ACC's 0.730 mm, Table III). Its push margin is, if anything, slightly better without the boost ($F_{\min} = 84.2$ N vs. 82.8 N), so the boost buys sagittal steadiness and settling, not capture basin. S5 (pitch gate) is critical to safety—removing it collapses $F_{\min}$ to 8.8 $\pm$ 0.6 N ($\sim 9\times$ below ACC) and additionally

Push Recovery Envelope ($N = 10$)

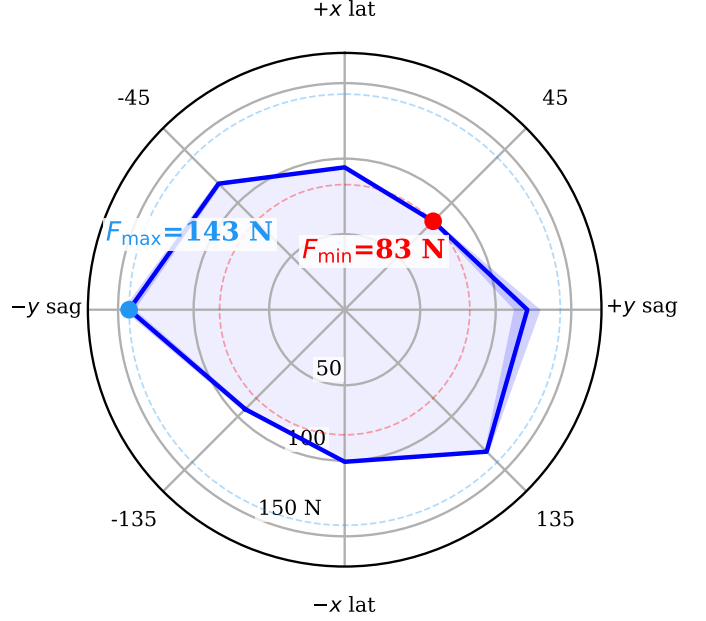


Fig. 4. Omnidirectional push envelope of ACC. This is the *same* run as row S1 of Table IV, not a separate measurement: 8 bearings, $N=10$ repetitions each, threshold located by binary search over 10–160 N to a 5 N tolerance, so the plotted minimum is by construction the $F_{\min}$ of that row. Solid curve: per-bearing mean survivable push magnitude; shaded band: ± 1 standard deviation over the ten repetitions. The radial axis is linear in force and originates at 0 N, so radial distance is proportional to margin. The envelope is strongly anisotropic, spanning 82.8–142.7 N—a 1.7 $\times$ ratio between the weakest and strongest bearing. The ordering is -90° (142.7 N) $>$ 135° (132.9 N) $>$ 90° (120.9 N) $>$ -45° (118.0 N) $>$ 180° (100.7 N) $>$ 0° (94.3 N) $>$ -135° (93.1 N) $>$ 45° (82.8 N). Two regularities are visible. First, the actuated axis is the strong one: bearings are applied in the world frame as $\mathbf{f} = F(\cos \theta, \sin \theta, 0)$, so $\pm 90^\circ$ are purely sagittal and $0^\circ/180^\circ$ purely lateral (Sec. V-A). The two sagittal bearings average 131.8 N against 97.5 N for the two lateral ones, a 1.35 $\times$ advantage. This is the expected asymmetry for a wheeled biped: sagittally the wheels accelerate the contact patch back under the CoM and convert the impulse into a controlled excursion, whereas laterally no actuator produces ground force at all and the only restoring authority is the static $w_{\text{track}} = 0.278$ m track width plus hip-roll PD. We compare axis *means* rather than ranks because the raw ordering is contaminated by the second regularity. Second, the envelope is measurably *chiral*: 135° survives 40 N more than -135° , and -90° 22 N more than 90° . An independent earlier sweep of a neighbouring configuration reproduces both asymmetries with the same sign and comparable magnitude, so this is a systematic property of the plant-plus-controller rather than sampling noise; we have not isolated its origin, and it is reported here as an observation rather than an explained effect (Sec. VI). $F_{\min}$ (red) is the bearing-wise minimum and is the figure of merit quoted throughout the paper, because it is the magnitude survived in *every* direction; $F_{\max}$ (blue) is shown only to indicate the spread.

causes 10/10 falls at pure idle. S3 (asymmetric EMA) and S2 are each individually required for ringdown.

C. Flight Recovery and Terrain Adaptation

ACC's two-channel assembly supports independent extensions without modifying the core anchor. Both extensions use the same 400 Nm/s rate limit and 100 Hz control rate.

Contact-loss recovery. ACC recovers to a settled stand from a 100 cm free drop (10/10, 24.0 $\pm$ 1.2 $^\circ$ peak pitch after touch-down) and from a 50 cm ledge drive-off (10/10, 27.1 $\pm$ 2.0 $^\circ$ peak landing pitch); the 60 cm drop and 20 cm ledge are

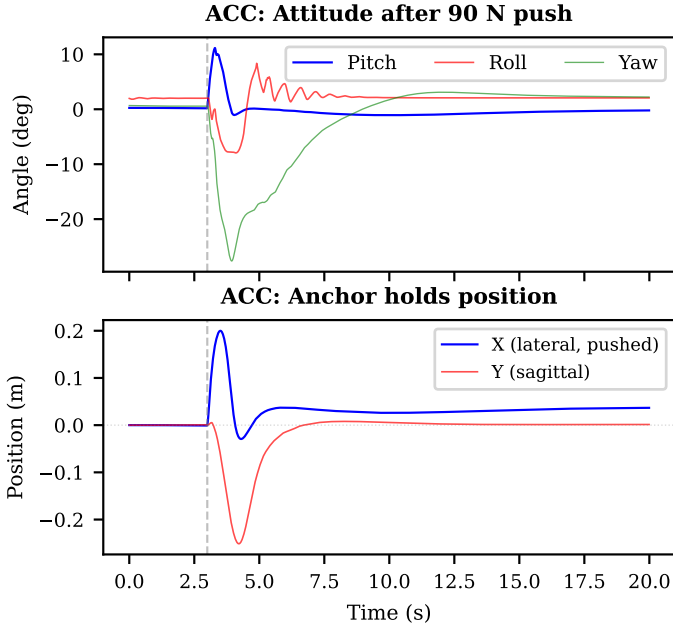


Fig. 5. Post-push recovery (90 N lateral, world +x, at $t=3$ s). **Top:** attitude angles; stabilization within 2 s. **Bottom:** planar CoM position, which separates the two axes cleanly. The lateral push couples through roll into a large *sagittal* excursion (-251 mm peak); the anchor drives that axis back to 1.37 mm of home and holds it there at 0.018 mm SD over the final 2 s—a 99.5% recovery. On the pushed lateral axis, which has no actuator authority (Sec. V-A), the 200 mm peak excursion decays only to a permanent 36.2 mm offset, 18% of the peak. The contrast between the two traces is the clearest single illustration of what the anchor does and where it does not act.

correspondingly milder (Table V). The two flight ablations, however, give a more nuanced picture than the mechanism’s design rationale suggests, and we report it as measured.

Removing the flight *attitude* PD (the reaction-flywheel term $2.0\theta + 0.5\dot{\theta}$, spin-down damping retained) is a real but modest loss: peak pitch rises from 24.0 to 27.8° on the 100 cm drop with one fall in ten, and from 27.1 to 31.6° on the 50 cm ledge with no falls. Removing the airborne wheel-torque override *entirely*—so the ordinary wheel-inverted-pendulum (WIP) loop keeps driving the wheels through flight—does not degrade either condition at all: it *lowers* peak pitch to $13.0 \pm 0.5^\circ$ (drop) and $10.2 \pm 0.4^\circ$ (ledge) at 10/10 survival, because free wheels driven by the pitch-feedback loop are themselves an effective reaction flywheel. The override’s benefit appears only at the upper end of the envelope: at 150 cm the two configurations survive equally (2/5 settled either way) but the override holds peak pitch to $34.0 \pm 2.1^\circ$ against $53.6 \pm 35.0^\circ$ without it—a $16\times$ reduction in spread, with one uncontrolled trial exceeding 90° . At 200 cm both configurations fail. The correct claim is therefore that the flight override **bounds landing-attitude variance near the envelope limit**, not that it enables the recovery; at the 50–100 cm scales reported here ACC’s balance core alone suffices. We state this explicitly because the mechanism was originally introduced to prevent a nose-down crash observed in an earlier controller revision, and that failure mode is no longer reachable in the present design.

Per-leg terrain adaptation. Here the ablation is unambiguous. With per-leg adaptation, straddle roll on the one-wheel curb is $2.5 \pm 0.04^\circ$ (10 cm), $3.5 \pm 3.2^\circ$ (15 cm) and $14.9 \pm 0.1^\circ$ (20 cm), all 10/10. Replacing `split()` with a symmetric posture (both legs commanded to the same height—the pre-adaptor behaviour) causes a fall at *every* tested curb height, 0/10 in all three cases: the torso rolls past 60° within 0.8 s of curb entry and the robot is on the ground by 2.3 s. The height split is thus load-bearing for this task in a way the flight override is not. The closed-loop leveling trim that rides on top of the split is a second-order refinement of the same mechanism: disabling it alone leaves the robot upright on the 10 cm curb ($3.2 \pm 0.08^\circ$, 10/10) but degrades the 15 cm curb to $11.1 \pm 1.6^\circ$ roll with 6/10 settled—the trim corrects the residual error of the non-linear height→leg-length table, which grows with step height. At 20 cm the limit is one of joint authority, not of the split geometry. Evaluating (21) at $h^{\text{cmd}}=0.404$ m gives commanded leg lengths of 0.350 m (flat) and 0.150 m (curb), i.e. a difference equal to the full 20 cm step with *zero* kinematic residual—the posture pair can span the curb exactly. What tightens is the joint budget: the compressed leg’s commanded knee angle grows 2.18 rad (10 cm) → 2.39 rad (15 cm) → 2.58 rad (20 cm) against a hard limit of 2.70 rad, leaving only 0.12 rad (6.6°) of headroom, and the commanded CoM height of that leg, 0.265 m, sits within 1.1 cm of the extrapolation floor $z_{10}-10\text{ cm}=0.254$ m. Because the commanded posture is exact, the measured 14.9° straddle roll is entirely *tracking* error: in this deeply flexed configuration the leg carries its load at the shortest gravitational moment arm the PID gains were never re-tuned for, and it settles below its target. Adaptation therefore still prevents leg-fold (10/10 survival), but the torso is left tilted; the tight $\pm 0.1^\circ$ std reflects a deterministic, saturation-dominated equilibrium rather than genuinely low run-to-run variability.

The gate structure isolates each extension to its physical regime without re-tuning the anchor, and both extensions cost three lines of controller code each. Their measured *necessity*, however, differs sharply: the per-leg height split is required for the curb task at every tested height, whereas the flight override is a variance-bounding refinement that only pays at the top of the drop envelope.

D. Comparison with Classical and Learning-Based Baselines

To establish ACC is not merely a tuned linear controller or un-structured policy, we compare seven baselines—six classical (four 4-state TWIP variants: LQR, LQR+AW, LQR-DT, PI+AW; two 6-state coupled variants: servo and direct-torque) plus a preliminary PPO baseline (all share identical environment setups):

- 1) **LQR**—4-state TWIP with decoupled roll/yaw PD through a PID servo layer (~ 0.25 s lag).
- 2) **LQR + Integral AW**—Same as (1) plus pitch-error integral with conditional AW (clamped ± 3.0 rad/s).
- 3) **LQR (Direct Torque)**—Re-optimized for direct-torque plant ($\tau_s=0$, $k_{p,\text{roll}}=55.0$), outputting torques directly.

TABLE V

CONTACT-LOSS RECOVERY AND PER-LEG TERRAIN ADAPTATION, WITH THE CORRESPONDING MECHANISM ABLATIONS. ALL AT 400 Nm/s RATE LIMIT, 100 Hz CONTROL. SURVIVAL IS THE *strict settle* VERDICT (FLAT PITCH/ROLL WINDOW, TAIL HEIGHT ERROR ≤ 12 mm, RESIDUAL SPEED ≤ 0.05 m/s), NOT MERELY STAYING UPRIGHT.

Contact-Loss Recovery			
Scenario / ablation	N	Peak Pitch ($^\circ$)	Settled
Drop 100 cm	10	24.0 $\pm$ 1.2	10/10
Drop 60 cm	10	18.2 $\pm$ 1.0	10/10
Ledge 50 cm	10	27.1 $\pm$ 2.0	10/10
Ledge 20 cm	10	19.6 $\pm$ 0.3	10/10
–Attitude PD, drop 100 cm ^a	10	27.8 $\pm$ 0.3	9/10
–Attitude PD, ledge 50 cm ^a	10	31.6 $\pm$ 2.2	10/10
–Flight override, drop 100 cm ^b	10	13.0 $\pm$ 0.5	10/10
–Flight override, ledge 50 cm ^b	10	10.2 $\pm$ 0.4	10/10
Drop 150 cm (envelope edge)	5	34.0 $\pm$ 2.1	2/5
–Flight override, drop 150 cm ^b	5	53.6 $\pm$ 35.0	2/5

Per-Leg Terrain (One-Wheel Curb, 0.15 m/s)			
Configuration	N	Straddle Roll ($^\circ$)	Settled
Full adaptation, 10 cm	10	2.5 $\pm$ 0.04	10/10
Full adaptation, 15 cm	10	3.5 $\pm$ 3.2	10/10
Full adaptation, 20 cm	10	14.9 $\pm$ 0.1	10/10
–Leveling trim, 10 cm ^c	10	3.2 $\pm$ 0.08	10/10
–Leveling trim, 15 cm ^c	10	11.1 $\pm$ 1.6	6/10
–Height split, 10 cm ^d	10	fall	0/10
–Height split, 15 cm ^d	10	fall	0/10
–Height split, 20 cm ^d	10	fall	0/10

$\pm$ std over N trials with randomized ICs (± 0.003 rad joint, ± 1.5 mm CoM). Drop metric: peak $|\theta|$ from touchdown onward; ledge metric: peak $|\theta|$ through landing; curb metric: max $|\phi|$ while straddling the curb top. ^aReaction-flywheel attitude PD zeroed; airborne spin-down damping retained. ^bEntire airborne wheel-torque override bypassed—the ordinary WIP loop drives the wheels through flight. ^cClosed-loop leveling integrator disabled; per-leg height split retained. ^d`split()` forced to a symmetric posture (both legs at the same commanded height), i.e. pre-adaptor behaviour; roll diverges past 60° within 0.8 s of curb entry, so no straddle-roll statistic exists. Produced by `scripts/collect_flight_terrain_n10.py`; raw output `outputs/flight_terrain_n10/results.json`.

- 4) **Coupled 6-State 3D LQR**—Extends (1) with roll states ($\phi, \dot{\phi}$).
- 5) **PI + AW (Direct Torque)**—DT-LQR + integral AW on fwd. position (conditional, ± 15 cm gate); same leg/roll/yaw as DT-LQR.
- 6) **Pure PPO (Preliminary)**—Model-free RL policy trained from scratch for 5.4M environment steps (single seed, in BalanceEnv).
- 7) **Coupled 6-State LQR (Direct Torque)**—Same 6-state model as (4) but outputs direct torques (bypassing PID servo, ~ 0.25 s lag removed). Gains re-derived for the direct-torque plant ($K_{\text{pitch}}=113$, $K_{\text{roll}}=58$ Nm/rad); leg posture via PD position control.

Direct-torque LQR removes the servo path, eliminating both the PID gains and the measured actuation lag. PI+AW adds integral anti-windup on forward position to DT-LQR—the standard engineering fix for a DC position bias, with windup protection. The anti-windup logic is verified independently by

One-wheel curb (20 cm) — per-leg terrain adaptation

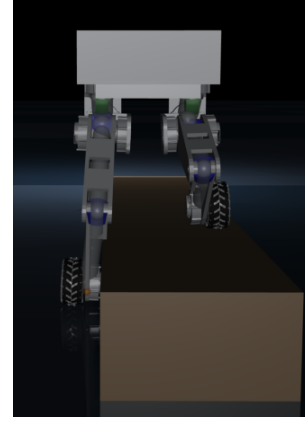


Fig. 6. One-wheel curb (20 cm). With the per-leg height split: 14.9 $\pm$ 0.1° straddle roll, 10/10 settled. With a symmetric posture (split removed): the curb-side leg cannot fold far enough, roll diverges past 60° within 0.8 s of curb entry, 0/10.

TABLE VI

BASLINE GAIN DERIVATION. ALL WEIGHTS FOLLOW BRYSON’S RULE; SEE THE NOTE BELOW THE TABLE.

Controller	Gains and derivation
LQR (4-state)	$Q = \text{diag}(10, 2, 3, 0.3)$, $R = 0.8$; roll and yaw closed by decoupled PD through the PID servo layer
LQR + AW	as above, plus a pitch-error integral with conditional anti-windup, clamped to ± 3.0 rad/s
LQR, DT	4-state gains re-optimized for the zero-lag plant: $\tau_s = 0$, $k_{p,\text{roll}} = 55.0$
6-state LQR	$Q = \text{diag}(10, 2, 3, 0.5, 3, 0.3)$, $R = \text{diag}(0.8, 1.0)$, derived in continuous time and shared by the servo and DT variants
6-state LQR, DT	state feedback re-solved for the zero-lag plant: $K_{\text{pitch}} = 113$, $K_{\text{roll}} = 58$ Nm/rad (see Limitation 7)
PI + AW, DT	DT-LQR gains plus an integral on forward position: $k_{i,\text{pos}} = 8.0$ rad/s/(m·s), clamp ± 5.0 rad/s, conditional gate ± 15 cm
PID servo layer	$k_p = [55, 40, 70, 70, 4]$, $k_d = [3, 2, 4, 4, 0]$, $k_i = [0.8, 0.4, 1, 1, 0.1]$, ordered (hip roll, hip yaw, hip pitch, knee, wheel)
Servo lag	0.25 s, measured as the wheel-velocity 90% rise time (≈ 25 steps at 100 Hz)

Bryson’s rule throughout ($Q_{ii} = 1/x_{i,\text{max}}^2$, $R = 1/u_{\text{max}}^2$); the 46-configuration retuning sweep described in the text confirmed that no retuning within these ranges changes the outcome. The 4-state model is (pitch, pitch-rate, forward velocity, forward position); the 6-state model adds (roll, roll-rate). “DT” denotes direct torque, i.e. the PID servo path removed and $\tau_s = 0$.

`scripts/validate_lqr_aw.py`. PPO evaluates what unconstrained model-free policy search achieves on the same plant. Every gain used for these seven baselines is listed in Table VI; we give them in full because the central claim of this subsection is a *negative* one—that no member of this family stands—and a negative result is only as strong as the tuning behind it.

Table VII reports the comparison. All six classical baselines—four derived from the 4-state TWIP model (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants

TABLE VII
LQR AND PPO BASELINES VS. ACC, CLEAN SENSORS. PARENTHESED
N IS THE EPISODE COUNT. SEE THE NOTE BELOW THE TABLE.

Configuration	Surv. (s)	θ_{RMS} (°)	ϕ_{RMS} (°)	H_{RMS} (mm)	τ_{RMS} (Nm)
LQR, TWIP ($N=60$)	0.52	0.04	24.2	371.0	92.10
LQR + AW ($N=60$)	0.52	0.04	24.2	371.0	92.10
LQR, DT (re-opt.) ($N=20$)	0.72	0.78	21.2	135.0	9.33
6-St. LQR ($N=20$)	0.52	0.04	24.2	371.0	92.10
PI + AW, DT ($N=20$)	0.72	0.78	21.2	135.0	9.33
6-St. LQR, DT ($N=20$)	0.16	0.00	21.1	117.0	13.90
Pure PPO (prelim.) ($N=20$)	3.20	4.82	3.15	54.2	18.40
ACC ($N=10$)	20.00	0.060	0.010	0.18	3.28

LQR / LQR+AW / 6-St. use the PID servo; LQR (DT), PI+AW (DT) and 6-St. (DT) command torque directly; PPO is a 5.4M-step RL policy (preliminary, single seed). The identical LQR / LQR+AW / 6-St. rows arise because all three saturate at the common 30 Nm limit through the same PID servo within 0.52 s, masking the gain differential.

Between-episode standard deviations (omitted above for legibility): LQR / LQR+AW / 6-St.: ± 0.00 , ± 0.01 , ± 2.1 , ± 45 , ± 3.2 ; LQR-DT / PI+AW-DT: ± 0.00 , ± 0.42 , ± 0.3 , ± 41 , ± 1.18 ; 6-St. DT: ± 0.00 on every metric (deterministic failure); PPO: ± 0.85 , ± 0.91 , ± 0.45 , ± 8.3 , ± 2.1 ; ACC: ± 0.00 , ± 0.006 , ± 0.001 , ± 0.02 , ± 0.00 .

All LQR and PI+AW variants fall within 0.72 s (100%); the 6-St. DT variant falls deterministically at 0.16 s (zero trial-to-trial variance, $N=20$), confirming that removing the servo path with gains derived from a simplified model does not rescue stability. PI+AW (DT-LQR + integral AW on fwd. position) yields metrics identical to DT-LQR: the integral cannot accumulate before roll-induced fall at ~ 0.72 s. Preliminary PPO achieves partial standing but suffers 85% falls over multi-height transitions ($H_{\text{RMS}}=54.2$ mm). $N=60$ for LQR/LQR+AW (sweep); $N=20$ for DT/6-St./PI+AW/6-St. DT/PPO.

(servo, direct-torque)—fall within 0.72 s. Removing the PID servo path (4-state DT-LQR: $\tau_s=0$, $k_{p,\text{roll}}=55.0$, no integral) reduces torque $10\times$ (92.1 Nm $\rightarrow$ 9.33 Nm) and marginally improves roll ($\phi_{\text{RMS}}=21.2^\circ$ vs. 24.2°). Adding a forward-position integral (PI+AW: $k_{i,\text{pos}}=8.0$, conditional ± 15 cm gate) changes nothing—the integral cannot accumulate before the roll-induced fall at 0.72 s—confirming the 4-state TWIP model is structurally insufficient for this articulated-leg platform. Preliminary PPO (5.4M steps, single seed) sustains single-height standing longer (3.20 s) but exhibits an 85% fall rate across multi-height command transitions (54.2 mm height error). ACC achieves $0.060\pm 0.006^\circ$ pitch RMS, $0.010\pm 0.001^\circ$ roll RMS, 0.18 ± 0.02 mm CoM Z, and 3.28 ± 0.00 Nm torque RMS ($N=10$). A **46-configuration** sweep (dimensions as given in Table VI; script `scripts/sweep_lqr_roll_gains.py`) confirmed all tested LQR variants fail (100%); a 6-state coupled LQR with direct torque actuation, removing the PID servo path, also fails—indicating platform complexity, not servo lag alone, is the dominant factor (see Limitation 7). Neither servo-lag removal nor preliminary PPO training (5.4M steps, single seed) resolved the height-adaptive posture trade-off in our experiments; whether a fully converged PPO policy closes this gap is not tested here.

One confound in this comparison is not resolved and we state it plainly rather than argue around it: ACC runs at 100 Hz and every baseline at 50 Hz, so ACC enjoys a $2\times$ control-bandwidth advantage. We attempted to remove it by re-running ACC at 50 Hz with the discrete EMA coefficients recomputed

for the longer step, and the attempt failed—the controller does not survive the rate change, and the resulting numbers are artifacts of a fallen robot rather than a rate-independence result. The full analysis is given in §VI (Limitation 5); the consequence for Table VII is that it establishes the classical baselines fail at the rate for which they were designed, and does *not* establish that they would still fail at 100 Hz. The gap between 20.0 s survival and 0.72 s is far larger than a factor-of-two bandwidth change plausibly accounts for, but we have not measured that, and we do not claim it.

E. Whole-Body Control Baselines

To address whether a tuned whole-body controller (WBC) could match ACC, we evaluated two WBC configurations that were developed and subsequently discarded during controller development: **(W1)** a task-priority quadratic-program (QP) WBC—minimizing COM height error, torso orientation error, posture damping, and wheel regularization subject to full-body dynamics and contact constraints—used as the *primary* balance controller; and **(W2)** the same WBC blended as an *assist* on top of ACC via $\tau_{\text{assist}} = \tau_{\text{ACC}} + \alpha \cdot (\tau_{\text{wbc}} - \tau_{\text{ACC}})$, with a seven-term multiplicative gate $\alpha_j = \alpha_{\text{max}} \cdot g_{\text{stability}} \cdot g_{\text{height}} \cdot g_{\text{push}} \cdot g_{\text{divergence}} \cdot A_j \cdot K_{\text{role},j}$ that continuously modulates per-joint WBC authority. Both were evaluated under identical cloned simulation conditions (three-arm counterfactual: ACC baseline, WBC-only, ACC+WBC assist; 225 scenarios spanning fixed-height balance, height transitions, and push recovery at 20–120 N; ACC profile K2_JAX_DEDICATED_DEFAULT_V3; no gain retuning).

W1 (WBC as primary controller). The standalone QP-WBC fails catastrophically: **17.2 $\times$** more falls than ACC (5,985 vs. 347 across 225 scenarios), **28.4 $\times$** more single-push falls (3,353 vs. 118), and pitch RMS 41.8° vs. ACC’s 15.9° ($2.63\times$).¹ Survival time at 0.60 m is 0.52 s (100% fall rate). The failure mechanism is structural: the WBC’s posture task is a velocity damper ($\dot{\mathbf{q}}_{\text{joints}} = -k_{\text{posture}} \cdot (\mathbf{q} - \mathbf{q}_{\text{current}})$, $k_{\text{posture}}=10$), not a position tracker—it cannot hold a target posture. It lacks gravity-compensation feedforward, uses a single linearization at 0.67 m base-z (vs. the 0.45–0.60 m base-z operating range of these runs, i.e. 0.32–0.47 m CoM-z; §III-A), and treats wheels as point contacts without a rolling constraint. The QP solver fails on 14.2% of steps, creating torque discontinuities. Four critical implementation bugs were identified post-hoc (contact Jacobian zero-gradient, solver feasibility-only hardcoded, base-z vs. CoM-z height reference mismatch, and gate height-reference conflict; detailed in Appendix B). WBC as-implemented-and-tuned therefore fails as a standalone controller. Critically, its failure is not solely a tuning or implementation defect: the WBC lacks the *regime-gating* that ACC’s proximity-gated anchor provides. A WBC distributes torque optimally for a fixed task set—it cannot express qualitatively distinct physical regimes (quiet stance,

¹The 15.9° here is ACC’s pitch RMS averaged over the mixed 225-scenario batch (including pushes and height transitions), not the clean standing value of 0.060° reported in Table VII.

push ringdown, large excursion) as distinct control modes. The precision–recovery trade-off ACC resolves through spatial gating and asymmetric temporal envelope-following is not expressible as a task-priority QP objective; even a bug-free WBC would lack the structured prior that gates the anchor integral, and would therefore face the same global-integral windup that our ablation S4 (no proximity gate, Table IV) confirms is fatal for sub-millimeter precision.

W2 (WBC as assist on ACC). The assist gate—designed to be maximally conservative—blocks WBC contribution in $\geq 99\%$ of steps: in the 225-scenario batch, all metric ratios are **1.000 000** (ACC+WBC assist $\equiv$ ACC within measurement noise); a 48-scenario quick evaluation classifies all 48 scenarios as ASSIST_EQUIVALENT (zero regressions, zero safety violations). This zero-net effect is *independent* of the WBC implementation bugs: the gate blocks WBC *before* its torque is computed, based on physical state (pitch, roll, height deviation, push magnitude, divergence), so the WBC’s internal correctness is irrelevant to the assist outcome.

When gates are manually overridden to force WBC contribution (keyframe test at 0.53 m base- z , $\alpha_{\max}=0.50$, widened stability and height thresholds), the assist produces mixed results: pitch RMS improves 17% ($2.44^\circ \rightarrow 2.04^\circ$), drift improves 18% ($0.070\text{ m} \rightarrow 0.058\text{ m}$), and yaw drift improves 20%—but roll RMS *degrades* 35% ($0.20^\circ \rightarrow 0.27^\circ$). The degradation arises from a fundamental architectural conflict: ACC’s lateral roll controller and WBC’s torso-orientation task both write to hip-roll joints, but from incompatible philosophies (decentralized PD vs. centralized QP exploiting full mass-matrix coupling). A posture-guided variant (WBC outputs posture targets $\mathbf{q}_{\text{ref}}$, ACC executes torque) reduced the roll degradation from +35% to +7% by eliminating torque-level interference, but did not produce a net improvement over ACC alone.

Timing overhead. The WBC QP solve alone consumes 0.45 ms mean, 2.94 ms p95, and **3.43 ms p99** (OSQP, offline Python/NumPy path, 4 scenarios). The p99 solve time alone exceeds the 0.5 ms end-to-end budget margin below the 10 ms performance threshold by 6.9 $\times$ —before accounting for matrix assembly, contact Jacobian computation, or the ACC computation baseline. An incremental JAX-JIT path was prototyped to reduce per-step overhead but was never integrated into the real-time control loop; real-time WBC assist overhead therefore remains uncharacterized. Even if the solve time could be reduced below the margin, the assist adds no measurable balance benefit (W2 zero-net result), so the compute cost is pure overhead.

Summary. Neither standalone WBC nor WBC-as-assist resolves the precision–recovery trade-off on this platform. The trade-off is resolved by regime-gating—the proximity-gated anchor and asymmetric envelope follower—which WBC’s task-priority formulation does not express. ACC’s conditional gates therefore provide the essential structured prior; the WBC gap is one of structural expressiveness, not torque optimality.

F. Architectural Analysis

We evaluate ACC under four dimensions: noise/delay robustness (Table VIII), gate parameter sensitivity, torque rate-limit invariance, and compound-disturbance recovery.

Robustness to sensor noise and actuator delay. A factorial sweep over four noise levels and three delays ($N=5$ per cell), extended along the clean-delay axis to 150 ms, reveals two distinct delay regimes: a **performance threshold** where $F_{\max}$ bifurcates (94–96 N at 0 ms vs. 41–50 N at any delay, idle RMS 5.5 $\times$), and a **stability threshold** above 150 ms—the robot survives all tested clean delays (10–150 ms, $N=5$, zero falls) with stable post-bifurcation performance ($F_{\max} \approx 49\text{ N}$, idle $\approx 2.5\text{ mm}$). ACC’s gates depend on physical state (position, velocity envelope, orientation), not tight timing; jitter that crosses the 10 ms performance threshold degrades push margin temporarily but does not cause a fall. High noise causes fall regardless of delay; the asymmetric EMA cannot distinguish noise from motion, forcing anchor disengagement. Idle repeatability degrades from 0.42 mm (clean) to 34.00 mm under medium sensor noise—an 81-fold loss. The comparison that makes this concrete is against the un-anchored baseline on the same axis: L0 holds $R_{\text{lat}} = 3.675\text{ mm}$ (Table III), so under medium sensor noise ACC is roughly an order of magnitude *worse* than having no anchor at all. The mechanism is the one described above—the envelope gate cannot separate sensor noise from real motion, so it releases the integral and then re-engages it repeatedly—and the practical consequence is that ACC as presented requires a filtered velocity estimate, not a raw one. Two caveats attach to the absolute values in this table. First, its idle column is measured on the *lateral* axis (world x), not the sagittal one of Table III, so the two are not directly comparable in level; the like-for-like clean reference here is ACC’s lateral $R_{\text{lat}} = 0.730\text{ mm}$, and the 0.42 mm entry is lower still because of the shorter settling allowance (3 s here vs. 5 s there): with $\text{EMA}_0=0$ the envelope gate opens fully at the start of the trial, so a window that begins earlier captures more of the strongly-anchored transient and less of the steady state. Second, the noise and delay *ratios* are unaffected by either issue, since every cell in the table is measured identically; it is the degradation factors, not the absolute millimeters, that this table is used for. Re-running the sweep with sagittal resolution and $\text{EMA}_0=1.0$ initialization would remove both caveats and is deferred.

Gate parameter sensitivity and tuning. Finite-difference gradients reveal a stark hierarchy: the **release coefficient** α_r **dominates** (sensitivity 220.4, 20 $\times$ the next, 130 $\times$ pitch gates); pitch stability thresholds (≈ 0) are effectively insensitive safety interlocks. For porting to a new morphology: (1) identify dominant ringdown period T_r from post-push decay; (2) set $\tau_{\text{release}} \approx 3\text{--}5 \times T_r$; (3) set quiet-stance band 2–3 $\times$ idle velocity noise floor; (4) set pitch gates conservatively ($\pm 12^\circ$, $\pm 15^\circ/\text{s}$); (5) tune k_p (fixed $k_p=50$ sufficed). A fixed k_p and anchor integral gain K_i require minimal adjustment across platforms; the critical parameter is τ_{release} .

Gate failure modes. Stuck-at-1 ($g_{\text{anchor}} \equiv 1$) causes windup $\rightarrow$

TABLE VIII

ROBUSTNESS TO SENSOR NOISE AND ACTUATOR DELAY. IDLE RMS: LATERAL (WORLD x) CoM STD. OVER 20 s AFTER 3 s SETTLE (EMA₀=0 TRANSIENT), $N=5$. $F_{\max}$: BINARY-SEARCH LATERAL (WORLD $+x$) PUSH, ± 5 N TOLERANCE, $N=5$.

Condition	Idle RMS (mm)		$F_{\max}$ (N)	
	Mean	$\pm$ Std	Mean	$\pm$ Std
Clean, 0 ms delay	0.42	0.03	96	1
Clean, 10 ms delay	2.30	0.16	47	2
Clean, 30 ms delay	2.62	0.22	46	3
Clean, 50 ms delay	2.61	0.42	49	8
Clean, 100 ms delay	2.23	0.25	48	4
Clean, 150 ms delay	2.40	0.22	45	1
Low noise, 0 ms delay	6.33	2.57	96	1
Low noise, 10 ms delay	12.82	3.93	47	2
Low noise, 30 ms delay	7.31	2.51	50	1
Med noise, 0 ms delay	34.00	11.28	94	3
Med noise, 10 ms delay	41.66	23.24	45	10
Med noise, 30 ms delay	35.10	16.83	41	10
High noise, 0 ms delay	3.00	0.55	95	3
High noise, 10 ms delay	3.00	0.55	95	3
High noise, 30 ms delay	— (fell)	— (fell)	— (fell)	— (fell)

Noise ($1-\sigma$): low = 0.01 rad/s (ang. vel.), 0.01 m/s² (grav.), 0.001 rad (joint pos.); med = 5 $\times$ low; high = 10 $\times$ low. Two distinct delay regimes: **performance** bifurcation at ~ 10 ms ($F_{\max}$: 96 N $\rightarrow$ ~ 47 N); **stability** threshold > 150 ms (zero falls at all tested clean delays, $N=5$; post-bifurcation idle ≈ 2.5 mm, $F_{\max} \approx 45-49$ N). The apparent medium $\rightarrow$ high noise improvement (34 mm $\rightarrow$ 3 mm) occurs because high noise holds EMA above threshold, disengaging the anchor into P-only mode. Non-monotonic low-noise RMS (12.82 mm at 10 ms vs. 7.31 mm at 30 ms) reflects resonance: 10 ms phase-shifts the anchor into intermittent engage/release; at 30 ms EMA stays above threshold (medium noise follows the same pattern).

Produced by scripts/collect_robustness_sweep.py (noise $\times$ delay grid, raw output outputs/paper_statistics/robustness_sweep.json) and scripts/collect_delay_stability_sweep.py (clean-delay stability threshold, raw output outputs/paper_statistics/delay_stability_sweep.json).

fall (cf. S4); stuck-at-0 yields P-only limit cycle (cf. L0); stuck-half-open triggers parametric resonance (cf. S5, $F_{\min}=9$ N). All three map to measured ablation rows (Table IV), providing implicit FMEA coverage.

Torque rate-limit invariance. The 400 Nm/s slew limit used throughout is not what sets ACC’s push ceiling. Sweeping it over an 8 $\times$ span, with each trial drawing an independent initial-posture perturbation (the same $\mathcal{N}(0, 0.005$ rad) joint / $\mathcal{N}(0, 0.001$ m) root draw as the push protocol of §V-B, $N=5$ per setting), gives lateral $F_{\max}$ of 92.5 ± 1.1 N (100 Nm/s) and 94.4 ± 0.0 N at 200, 400 and 800 Nm/s, with idle lateral RMS of 0.695 ± 0.046 , 0.738 ± 0.031 , 0.732 ± 0.025 and 0.729 ± 0.023 mm respectively—a 2% spread in push margin and a statistically indistinguishable spread in idle precision. The limiter is genuinely active where it could matter: instrumenting the commanded torque during a 90 N push shows $|\dot{\tau}|$ saturating exactly at each configured bound (100, 200, 400 Nm/s; at 800 Nm/s the wheel channel still reaches the bound while the leg channels peak at 712 Nm/s). During quiet stance, by contrast, the peak commanded slew measured across the four settings is 0.51–1.66 Nm/s—60–190 times below even the 100 Nm/s bound—so idle precision cannot

TABLE IX

COMPOUND DISTURBANCE: LATERAL (WORLD $+x$) PUSH DELIVERED AT THE MIDPOINT OF A 1 s COMMANDED HEIGHT RAMP Δh ABOUT THE 0.404 m NOMINAL CoM- z . $N=10$ PER ROW, INDEPENDENT INITIAL-POSTURE DRAWS, BINARY SEARCH OVER [10, 130] N. $\Delta h=0$ IS THE MATCHED STATIC CONTROL. ROWS WITHIN ± 5 cm STAY INSIDE THE CALIBRATED POSTURE BAND (0.354–0.454 m); THE ± 10 cm ROWS ARE EXTRAPOLATED.

Δh	Band	$F_{\max}$ (N)	vs. static	Peak pitch (°)
+10 cm	extrap.	18.2 ± 5.2	–81 %	9.9 ± 14.0
+5 cm	calib.	92.9 ± 0.8	–2.0 %	15.9 ± 0.5
0 (static)	—	94.8 ± 0.8	—	14.0 ± 0.2
–5 cm	calib.	96.2 ± 0.0	+1.5 %	11.9 ± 0.5
–10 cm	extrap.	80.7 ± 6.0	–15 %	12.1 ± 2.3

Mean $\pm$ SD. All differences from the static control are significant under Welch’s t -test at the Bonferroni-adjusted $\alpha=0.007$ of §VI ($p < 4 \times 10^{-4}$ for the ± 5 cm rows, $p < 4 \times 10^{-5}$ for the extrapolated rows). Peak pitch is measured over the whole episode at each trial’s own $F_{\max}$, so it is a shape statistic at the survival boundary, not a severity ranking across rows. Produced by scripts/compound_disturbance_corrected.py; raw output outputs/compound_disturbance/results_corrected.json.

be slew-limited at any tested setting. The ceiling is therefore architectural rather than actuator-limited.²

Compound-disturbance recovery. The push protocol of §V-B holds the height command fixed. Because the posture map is height-scheduled, a commanded transition and a push compete for the same leg-torque budget, so the two disturbances are evaluated together here. Scenario A applies the standard lateral push impulse at the midpoint of a 1 s commanded height ramp of size Δh about the 0.404 m nominal CoM- z , and locates $F_{\max}$ by the same binary search; $\Delta h=0$ is the matched static control, run under the identical protocol so that the two differ only in the height command. Every trial draws an independent initial posture from the §V-B distribution, $N=10$ throughout. Table IX reports the result.³

Three findings follow, and they change the earlier reading of this experiment. First, *inside the calibrated posture band a simultaneous height command is nearly free*: a +5 cm rise costs 2.0% of push margin (92.9 N vs. 94.8 N) and a –5 cm squat is marginally *better* than static (96.2 N, +1.5%), because squatting lowers the CoM and shortens the toppling moment arm while the push arrives. Both effects are statistically resolv-

²An earlier version of this sweep reported invariance over 200–800 Nm/s from six bit-identical rows. Both features were artifacts: the sweep wrote the bound in Nm/s divided by the control step, overshooting by 100 $\times$ so that the limiter never bound, and it accepted a per-trial seed that it never used, so the five “trials” at each setting were the same deterministic run. The numbers above come from the corrected sweep (scripts/sweep_rate_limit_corrected.py, raw output outputs/rate_limit_sweep/results_corrected.json) and supersede it; the slew instrumentation is scripts/probe_torque_slew.py.

³An earlier version of this experiment reported a single 84% degradation, described as a “0.65 m $\rightarrow$ 0.50 m squat.” Two defects invalidated it. The script in fact ramped the command *upward* from the 0.404 m nominal to 0.50 m—a ~ 10 cm *rise* into the extrapolated region, not a squat—and the label reflected an earlier base- z convention (§III-A). It also accepted a per-trial index that it never used, so the five “trials” were one deterministic run repeated and the reported spread was structurally zero. The corrected experiment supersedes it. Its $\Delta h=+10$ cm row reproduces the collapse, confirming the effect was real; what was wrong was its sign, its magnitude, and the claim that it applies to height transitions in general.

able but operationally negligible, and they bracket the static value—there is no shared-budget penalty at this amplitude. Second, the degradation is confined to the *extrapolated* region and is strongly asymmetric: a +10 cm rise collapses $F_{\max}$ to 18.2 N (−81 %), whereas the mirror-image −10 cm squat costs only 15 %. The asymmetry identifies the mechanism. Rising drives the knee toward extension, where the leg Jacobian is worst conditioned and the same joint torque produces the least vertical force, and it does so while raising the CoM; squatting moves in the opposite direction on both counts. The failure is therefore a property of the posture map’s extrapolated upper end (§III-A), not of the anchor, and the operational consequence is a commanded-range recommendation rather than an architectural limit: transitions within the calibrated band are safe under push, transitions above it are not. The large peak-pitch spread of the +10 cm row ($\pm 14.0^\circ$, one trial reaching 48.9°) is consistent with trials sitting on the edge of the capture basin rather than comfortably inside it. Third, Scenario B tests sequential rather than simultaneous disturbance: a 90 N lateral impulse followed 2 s later by a 60 N counter-directed impulse, before the first ringdown has fully decayed. ACC survives **10/10** (Clopper–Pearson 95%CI [0.69, 1.00]) with an episode peak pitch of $11.89 \pm 0.51^\circ$, essentially all of it from the first impulse: after the reversal the pitch peaks at only $2.78 \pm 0.11^\circ$ and never re-enters the 5° band used as the settle criterion, in every trial. The reversal is thus absorbed within the residual motion of the first recovery, which is the behavior the asymmetric envelope is designed to produce— $\tau_{\text{release}} \approx 1.5$ s keeps the damping boost still partly engaged when the second impulse arrives.

VI. DISCUSSION AND CONCLUSION

ACC achieves all demonstrated behaviors without neural network training, is hypothesized to provide a *reduced* sim-to-real transfer gap vs. learned policies—pending hardware validation [22]. Each gate was added after tracing a measured failure mode to its root cause: (i) bistable envelope; (ii) global-I windup; (iii) parametric pumping; (iv) hard-leash oscillation; (v) self-referential stiffness. All thresholds frozen prior to the ablation sweep (Table IV).

Statistical power. Welch’s t -test, Bonferroni-adj. $\alpha=0.007$ (7 comparisons); L0→L1 and L2→L3 significant ($p<0.001$, Cohen’s $d \approx 5.3, 6.4$). The primary idle claims are stated as like-for-like ratios in Sec. V-A ($59.4\times$ sagittal repeatability, $22.6\times$ planar repeatability, $5.0\times$ lateral repeatability, all L0 vs. ACC at $N=10$ under one protocol; the headline is the sagittal figure, which is the actuated axis); the S5 collapse is unambiguous (0/10 survival at idle, 10/10 at every push magnitude). Flight/terrain: $N=10$, Clopper–Pearson 95%CI [0.69, 1.00].

Limitations. (1) The push envelope is anisotropic and chiral. The weakest bearing is the oblique 45° (82.8 N, 29 % below F_{med}), and the two lateral bearings average 97.5 N against 131.8 N for the two sagittal ones—the wheels can drive the contact patch back under the CoM sagittally but produce no lateral ground force. The chirality (Fig. 4) is reproducible

but unexplained. (2) Oblique 45° ledge descent at ≥ 20 cm exceeds lateral balance authority. (3) All results are simulation-only (MuJoCo); hardware validation remains future work. (4) ACC exhibits two distinct delay thresholds: a **performance** bifurcation at ~ 10 ms ($F_{\max}$ drops ~ 95 N $\rightarrow$ ~ 45 N, the 0.5 ms end-to-end budget margin is for this threshold); and a **stability** threshold > 150 ms (robot survives all tested clean delays, zero falls, $N=5$; see Deployment Risks). A jitter spike exceeding the performance threshold degrades push recovery temporarily but does not cause a fall—ACC’s gates depend on physical state, not tight timing. (5) **The control-rate comparison is confounded and unresolved.** ACC runs at 100 Hz; the LQR and PPO baselines run at 50 Hz. A 50 Hz ACC evaluation with recomputed discrete coefficients ($\alpha_a \approx 0.58$, $\alpha_r \approx 0.013$, anchor leak 0.010) *failed*: the robot collapsed to a mean CoM height of 0.124 m (nominal 0.404 m), and the sub-millimeter idle RMS and 10 N push threshold recorded by that run are measurements taken on a fallen robot. We therefore cannot claim that ACC’s architectural contribution is rate-independent, and the $2\times$ bandwidth advantage over the baselines remains an unremoved confound in Table VII. Two observations bound how much it can explain: DT-LQR, which removes the PID servo path (the dominant lag source) and so is far less rate-sensitive, still falls within 0.72 s with no push applied; and the failure modes differ in kind (baselines diverge in roll within one second, whereas ACC stands for the full 20 s horizon), which a bandwidth factor alone does not obviously produce. Neither observation substitutes for a rate-matched comparison. Re-tuning ACC for the 20 ms step, and re-running the baselines at 100 Hz, are both required future work. (6) **The commanded height range is bounded by the posture map, not by the anchor.** Within the calibrated band (0.354–0.454 m CoM- z) a commanded transition costs at most 2 % of push margin, but a +10 cm rise into the extrapolated region collapses $F_{\max}$ by 81 % (Table IX); the mirror-image squat costs only 15 %. Operating above the calibrated band under disturbance is therefore unsafe, and extending the band requires additional posture calibration points rather than a controller change. ACC has not been evaluated over a wide commanded-height sweep in the manner of a dedicated height-tracking study; only the ± 5 cm and ± 10 cm steps of §V-F are measured. (7) Six classical baselines—four 4-state TWIP variants (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo and direct-torque)—all fail (100 %). A full-state LQR from finite-difference linearization of the complete 10-DOF MuJoCo plant was attempted but produced unreliable gain matrices due to ill-conditioned contact dynamics at the standing equilibrium ($\kappa(A) \approx 10^{10}$); robust linearization of articulated-leg contact dynamics at unstable equilibria is an open challenge. The hip-roll coupling $b_{r,h} = -5.0$ in the 6-state LQR is empirical; a sensitivity sweep is deferred. (8) **A 27 mm sagittal parking offset persists on the very axis the anchor regulates.** ACC’s position integral acts on the sagittal axis, yet the robot settles -27.0 mm from the commanded home along that axis and then holds the offset to within 0.009 mm SD across ten trials (Table III). Sagittal *repeatability* is excellent (0.294 mm), so

the deficit is a static bias rather than motion, and it is not a windup artifact: it is stationary, reproducible to a hundredth of a millimeter, and largely insensitive to the gate structure (every subtractive ablation that survives lands within 0.2 mm of it). The most likely cause is a mismatch between the latched home position and the true torque-balance equilibrium—the controller is holding the position at which the anchor integral’s output balances the standing pitch bias, not the position that was commanded—compounded by the CoM- z versus base- z convention gap (§III-A). Disabling the integral moves the offset the wrong way, to -31.2 mm (M1, Table III), confirming the integral removes 4.2 mm of it but not the bulk. Absolute positioning accuracy is therefore limited to ~ 27 mm and is the binding constraint on any task requiring it; this is the design’s principal open weakness. Calibrating the latched home against the measured equilibrium, and raising the anchor gain now that windup is gated, are the two direct routes to closing it. Both are untested. (9) Architectural generality across morphologies remains to be demonstrated. (10) Push ablation at $N=10$ has min. detectable effect ~ 3.2 N; sub-2 N differences non-significant (see Statistical power). The PPO baseline (5.4M steps, single seed) may not be fully converged; the 85 % fall rate is reported as observed, not as an upper bound on RL performance for this platform. A task-priority QP whole-body controller was benchmarked both standalone and as an assist on ACC and did not resolve the precision–recovery trade-off (§V-E); re-running with known implementation bugs fixed is deferred (the architectural argument—absence of regime-gating in a fixed-task-priority formulation—does not depend on it). A tuned convex MPC on the full 10-DOF plant remains deferred because finite-difference linearization at the standing equilibrium is ill-conditioned ($\kappa(A) \approx 10^{10}$, Limitation 7), so robust full-plant MPC is an open prerequisite rather than a missing comparison.

Deployment Risks and Mitigations. ACC uses raw torque, 400 Nm/s rate limit, standard sensors (IMU + encoders). We distinguish two latency thresholds: a **performance** threshold (~ 10 ms) where $F_{\max}$ bifurcates 96 N $\rightarrow$ ~ 47 N, and a **stability** threshold (> 150 ms, clean sensors, $N=5$, zero falls) where the robot actually falls. ACC’s gates (proximity, envelope, pitch) depend on physical state, not timing precision—the architecture is intrinsically delay-tolerant for stability. The 150 ms+ stability threshold is consistent across all physical gating mechanisms: the proximity gate (spatial, 15 cm), asymmetric envelope (temporal, $\tau_r \approx 1.5$ s), and damping boost are all state-driven rather than schedule-driven.

Estimated non-RTOS worst-case budget: sensor readout ~ 1 ms + state estimation ~ 2 ms + ACC computation ~ 3 ms + actuation latency ~ 3.5 ms = ≈ 9.5 ms end-to-end. The 0.5 ms margin below the 10 ms performance threshold means any single jitter source (OS scheduling preemption, bus contention) can degrade push recovery margin temporarily; the stability margin remains > 140 ms. On a real-time OS with a dedicated balance-loop microcontroller (e.g., 1 kHz RT Linux or bare-metal MCU), jitter is typically < 0.1 ms, recovering 2–4 ms of budget and making the performance margin comfortable.

Neither scenario has been validated on hardware—latency characterization with an oscilloscope and jitter histogram is required before deployment. The discarded WBC-assist variant (§V-E) is excluded from the hardware path not only because it adds no measurable balance benefit (225-scenario equivalence), but because its QP solve alone (p99 3.43 ms) exceeds the 0.5 ms margin below the 10 ms performance threshold by $6.9\times$ in the offline path; JAX-JIT incremental integration was prototyped but not validated, and the zero-net benefit makes further optimization moot. Hardware degradation: 0.294 mm sagittal repeatability $\rightarrow$ projected 1–3 mm (encoder quantization, backlash, stiction, tire compliance, IMU noise; cf. Table VIII). Staged commissioning: static torque $\rightarrow$ standing $\rightarrow$ push recovery $\rightarrow$ flight (tethered) $\rightarrow$ terrain.

We presented ACC, a two-channel torque assembly where a PD balance core is augmented by a proximity-gated anchor integral achieving 0.294 ± 0.015 mm idle repeatability on the sagittal (actuated) axis—a $59\times$ improvement over the proportional-only baseline under one unified protocol ($N=10$; Table III)—with omnidirectional push survival ($F_{\min}=83$ N, $F_{\text{med}}=117$ N). Absolute accuracy on that axis is an order of magnitude worse and we report it as such: the robot parks 27.0 mm from the commanded home, holds that offset to 0.009 mm, and closing it is the clearest remaining work on the design. Six classical baselines—four 4-state TWIP variants (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo and direct-torque)—all fail (100 %); adding integral action to LQR-optimal feedback does not prevent the fall, confirming ACC’s structured prior as the essential contribution. ACC tolerates moderate noise, is insensitive to the actuator slew bound over an $8\times$ span (100–800 Nm/s: 2 % spread in $F_{\max}$, no resolvable change in lateral idle precision), exhibits a push-performance delay bifurcation at ~ 10 ms with stability maintained beyond 150 ms, and handles contact loss and terrain without neural network training. Future work: hardware validation, EMA settle-time characterization, and extending conditional activation to locomotion and as a structured prior for residual RL.

CODE AND DATA AVAILABILITY

The controller, the MuJoCo model, every evaluation script referenced above, and the raw result files backing each table are available at <https://github.com/thuong180702/Wheeled-bipedal-robot-simulation> (branch `main`; a release tag will be minted at publication). Each main results table names its producing script and its raw result file in the note directly beneath it (Tables III, IV, V, VIII and IX); the remaining sweeps are cited inline at the point where they are discussed. The two superseded experiments of §V-F ship both the original and the corrected result files, so the correction itself is auditable.

APPENDIX A WHY SATURATION-TRIGGERED ANTI-WINDUP IS INSUFFICIENT

Saturation-triggered AW triggers on actuator saturation; on this platform, position error grows to 10–30 cm during a push while wheels stay within their 20 rad/s range, so saturation-triggered AW never activates. A static dead-zone either over-restricts or under-restricts; ACC’s proximity gate with continuous smoothstep avoids both. State-gated conditional integration is practiced in process control; ACC couples spatial gating with asymmetric envelope-detected damping, providing temporal gating absent in static-threshold schemes. AW-augmented baselines (LQR+AW, PI+AW) are evaluated in Table VII; broader AW taxonomy comparison is deferred.

APPENDIX B WBC BASELINE IMPLEMENTATION DETAILS

A. Standalone WBC (W1): Full Metric Table

Table X reports all measured metrics for the standalone QP-WBC (W1) vs. ACC, from the 225-scenario three-arm counterfactual evaluation and the 4-scenario Phase 3D full-batch execution. The WBC was evaluated with known implementation bugs (Table XI); W1 numbers may therefore be pessimistic. However, the failure mechanism—velocity-damping posture, single-linearization, no rolling model—is architectural, and the absence of regime-gating (the core ACC contribution) is structural, not bug-dependent.

TABLE X
STANDALONE QP-WBC (W1) VS. ACC — FULL METRIC COMPARISON

Metric	WBC-only	ACC (V3)	Ratio (WBC/ACC)
Total falls (225 scenarios)	5,985	347	17.2×
Single-push falls	3,353	118	28.4×
Falls (Phase 3D, 4 step_e scenarios)	14,713	0	—
Survival time at 0.60 m (s)	0.52	20.0	0.026×
Pitch RMS (°)	41.8	15.9	2.63×
Roll RMS (°)	96.8	84.2	1.15×
Yaw drift (°)	8.0	4.24	1.89×
Height RMS (m)	0.253	0.261	0.97×
Planar drift (m)	2.57	2.42	1.06×
Posture error (ratio vs. ACC)	—	—	8.05×
Yaw error (ratio vs. ACC)	—	—	15.43×
QP solver success rate	85.85 %	—	—

Sources: 225-scenario batch (see V3_vs_V3_Assist_comparison_report.md); Phase 3D (full_batch_verdict.json); survival time (best_torque_wbc_summary.json). All runs with ACC profile, no gain retuning. WBC evaluated with known bugs (Table XI); W1 numbers may be pessimistic.

B. Known WBC Implementation Bugs (F1–F4)

Table XI lists the four critical bugs identified during post-hoc audit of the WBC implementation. These bugs affect W1

(standalone) metrics; W2 (assist) zero-net is bug-independent because the gate blocks WBC torque before it is computed.

TABLE XI
WBC IMPLEMENTATION BUGS (POST-HOC AUDIT)

ID	Description	Effect on Metrics
F1	Contact Jacobian zero-gradient: leg joint columns = 0 (autodiff through cached FK).	Ground reaction forces produce zero moment on leg joints; τ_{wbc} meaningless at hip-pitch/knee ($K_{role}=0.60$).
F2	Fast solver hardcodes <code>feasibility_only</code> —drops entire task stack.	QP finds minimum-norm $\dot{q}$; all task modes produce identical output.
F3	Offline eval passes base-z (0.536 m) as height cmd. instead of CoM-z (0.404 m)—13.2 cm error.	ACC gain schedule and feedforward at wrong operating point; drift/heaving/centering gates disabled.
F4	Gate height conflict: WBC linearized at 0.67 m base-z while the gate compares against a 0.45–0.60 m base-z command band.	$g_{height} \approx 0$ at all test heights; WBC gate forced closed.

Source: docs/validation/v3_wbc_assist_root_cause_audit.md. F1–F4 affect W1 (standalone); architectural argument (absence of regime-gating) does not depend on them. W2 (assist) zero-net is bug-independent: gate blocks WBC before torque is computed.

C. W2 Zero-Net Result Is Bug-Independent

The assist gate α_j is computed from physical state (pitch, roll, height deviation, push magnitude, divergence) *before* WBC torque is evaluated. Since $\alpha_j \approx 0$ in $\geq 99\%$ of steps, the WBC’s internal correctness—buggy or not—is irrelevant to the assist outcome. The 225-scenario equivalence (ACC + WBC assist $\equiv$ ACC within measurement noise) is therefore a robust result. The keyframe test (gates manually overridden) reveals the torque-blend conflict (roll +35 %) independently of F1–F4: with the gate forced open, WBC torque is applied, and the decentralized-vs-centralized conflict manifests as roll degradation.

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